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Monte-Carlo HJ Reachability

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Backward reachability for differential games are computed from Hamilton–Jacobi equations whose numerical solution in the form of backward reachable tubes (BRTs) becomes costly as state dimension increases. We study a generalized Cole–Hopf transformation of the variational HJ equation employed in computing BRTs while retaining its residual. The transformed problem is a semilinear heat equation whose mild representation couples Gaussian initial-condition and nonlinear source integrals. We derive its spatial gradient, establish conditional finite-time convergence of source iteration on a positive invariant set, and obtain a residual certificate by comparison. Multidimensional verification combines exact solutions, direct integration of the full transformed equation, and independent viscosity references for control and pursuit–evasion problems. A starling-inspired four-component model admits supports evaluation at 100,000 4D states, with computations that assess discretization and population-query accuracy.

Slides Code

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1. Introduction

Modern complex systems — with their diverse interconnection of actuators, sensors, control systems, and the software modules that coordinate their various processes and events — require software verification tools capable of geometrically reasoning about their future behaviors. Given the continuous and/or hybrid systems characterization of their dynamics, simulation and testing, verification, observation, and further testing loop cycles are the standard norm for detecting unforeseen behavior ahead of deployment. Owing to their scale, however, analyzing system behavior based on closed-form analytic mathematical tools is often infeasible. This has stimulated efforts that complement traditional mathematical analysis tools from dynamical systems with emerging techniques in reachability [1], viability [2], or invariance [24] theories. These new tools allow exploring all possible scenarios where faults may arise for safety assurance; that is, ascertaining the freedom of a system from harm by generating evidence that a system, or any its components satisfy all specified requirements including functional and allocated baselines(i.e., *verification*) [15]. Examples of complex systems include smart manufacturing, ground [35] and air traffic management systems [55], flight control systems [41], or distributed systems.

Differential optimal control and game theories are useful formal verification tools for certifying the runtime safety assurance [24] of multiple agents interacting over a shared space in a *geometric* sense [28]. Both rely on a resolution of the Hamilton-Jacobi-Bellman (HJB) and/or the Hamilton-Jacobi-Isaacs (HJI) equation [13]. Being partial differential equations (PDEs), general high-dimensional HJB/HJI equations rarely admit closed-form solutions. Thus, approximation techniques have become the default mechanism for formal systems verification in these settings. Reachability theory [42, 55] is one such approximation technique. It entails computing the essentially non-oscillating (ENO) [49] Lax-Friedrichs [12] approximations to the HJ PDE via levelset methods [43, 45, 47]. Propagating the fronts of a conservation law on a mesh, the inviscid Euler equations' solution reduces to the derivative of the HJ problem's solution [18]; such that integrating this solution with total variation diminishing Runge-Kutta (TVD-RK) [48] approximates *viscosity solutions* to the original HJ problem.

HJ equation systems are employed within reachability analysis to evaluate a dynamical system's behavioral (or trajectory) evolution throughout a state space for example, given its initial point sets (states). The problem is typically setup so that the initial set must satisfy *input or state constraints* in a *least restrictive sense* and under a *worst-possible disturbance* [37]. Reachability is thus a useful geometric reasoning tool for determining the states that can be reached from an initial condition in a finite number of steps [42, 45] . *Reachability* can be analyzed in a (i) *forward* sense, where system trajectories are examined to determine if they enter certain states from an *initial set*; (ii) *backward* sense, where system trajectories are examined to determine if they enter certain *target sets*; (iii) *reach set* sense, in which they are examined to see if states reach a *set* at a *particular time*; or (iv) *reach tube* sense, in which they are examined to determine that they reach a set *during a time interval*.

This article studies the Cauchy equations associated with terminal reachability and backward reachable tubes. A terminal reachable set concerns target entry at a fixed horizon; a tube permits entry during an interval. Reach–avoid problems additionally constrain the trajectory before target entry and require an obstacle or stopping formulation. The retained-source Cauchy analysis does not supply that additional guarantee. Classical tensor-grid discretizations require storage that grows exponentially with the number of state coordinates at fixed resolution [5]; avoiding that storage does not alone establish numerical accuracy or favorable complexity.

Consider the evolution-form of the Cauchy-type HJ PDE

$$\boldsymbol{v}_t + \boldsymbol{H}(t;\boldsymbol{x}, D\boldsymbol{v}) = 0 \text{ in } \Omega \times (0,T], \quad \boldsymbol{v}(0;\boldsymbol{x}) = \boldsymbol{g}(0;\boldsymbol{x}) \text{ on } \Omega \times \{t=0\} \tag{HJ-IVP}$$

Here $\boldsymbol{x} \in \Omega \subseteq \mathbb{R}^n$, the Hamiltonian $\boldsymbol{H}$ is continuous, and $D\boldsymbol{v}$ is the spatial gradient. The initial condition $\boldsymbol{g}$ is bounded and uniformly continuous in the standing Cauchy theory. Crossing characteristics can destroy classical differentiability even when the Hamiltonian and initial condition are smooth [10]. Viscosity solutions provide the appropriate generalized solution

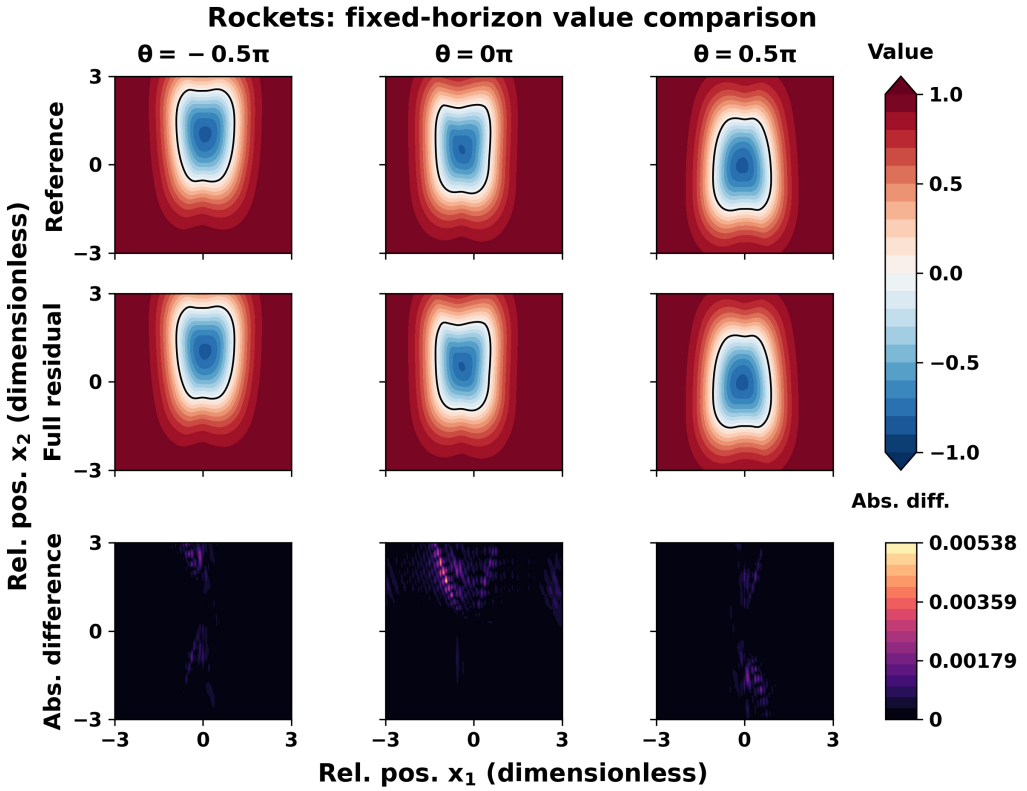


Figure 1: Normalized rockets benchmark at elapsed horizon $T = 0.5$, with $\delta = 0.08$. Rows show the independent finite-difference reference, direct integration of the complete transformed equation, and their absolute difference. Columns select $\theta = -\pi/2, 0$, and $\pi/2$. Both solvers use the same initial condition and viscosity. Black contours mark zero, and coordinates and payoff are dimensionless. These are fixed-horizon values; the numerical validation protocol is given in Section 3(b).

and comparison framework under the corresponding structural hypotheses. We study a fixed positive-viscosity regularization before considering its approximation error.

Retaining the residual of a generalized Cole–Hopf transformation yields a semilinear heat equation that is equivalent to the viscous Hamilton–Jacobi equation under positive classical logarithmic inversion. The source retains the Hamiltonian and coefficient derivatives. Gaussian kernels represent its linear subproblems, but their source depends on the unknown space–time value and gradient. This construction uses the semilinear parabolic framework of [23, 27]; the transformation alone does not imply a new dimension-independent complexity result.

The theoretical results concern three questions. First, the transformation identity and its differentiated mild representation retain every source term. Second, a short-time source iteration contracts on a positive invariant set under explicit value-and-gradient Lipschitz assumptions. Third, comparison converts a uniformly bounded original-equation defect into a value-error budget and conditional sign certificate. Finite-query concentration controls specified source evaluations; it does not by itself provide a continuum safety guarantee.

Numerical experiments integrate the complete transformed equation for control and pursuit–evasion problems and compare its fields with independent discretizations of the same viscous equation. Exact multidimensional manufactured solutions check the implementation and refinement behavior. A starling-inspired model connects these field computations to population-scale value queries through an explicit altitude-invariance reduction.

2. Background and Quasilinearization Framework

Backward reachability associates a differential game with a Hamilton–Jacobi–Isaacs equation [42]. Let $\mathcal{U} \subset \mathbb{R}^m$ and $\mathcal{W} \subset \mathbb{R}^p$ be compact control sets. At physical time $s \in [0,T]$, let $\overline{\mathcal{U}}(s)$ and $\overline{\mathcal{W}}(s)$ contain the measurable control signals on $[s,T]$. A minimizing strategy $\beta : \overline{\mathcal{U}}(s) \to \overline{\mathcal{W}}(s)$ is nonanticipative if equal input histories almost everywhere on $[s,r]$ produce equal output histories there. Denote the strategy class by $\mathcal{B}(s)$.

The trajectory ξ starts at $\boldsymbol{x}$ and satisfies

$$\dot{\xi}(r) = f(r, \xi(r), \boldsymbol{u}(r), \beta[\boldsymbol{u}](r)), \qquad \xi(s) = \boldsymbol{x}, \quad s \le r \le T. \tag{2.1}$$

Assume that f is bounded and continuous and uniformly Lipschitz in the state. For bounded uniformly continuous running cost l and terminal cost $\boldsymbol{g}$, the lower value is

$$\boldsymbol{v}_{\mathrm{phys}}(s,\boldsymbol{x}) = \inf_{\beta \in \mathcal{B}(s)} \sup_{\boldsymbol{u} \in \overline{\mathcal{U}}(s)} \left\{ \int_s^T l(r, \xi(r), \boldsymbol{u}(r), \beta[\boldsymbol{u}](r))\, dr + \boldsymbol{g}(\xi(T)) \right\}. \tag{2.2}$$

The first player maximizes this payoff and the strategy player minimizes it. Under the corresponding dynamic-programming and comparison hypotheses, the physical-time Hamiltonian is

$$\boldsymbol{H}_{\mathrm{phys}}(s,\boldsymbol{x},\boldsymbol{p}) = \max_{\boldsymbol{u} \in \mathcal{U}} \min_{\boldsymbol{w} \in \mathcal{W}} \{\boldsymbol{p} \cdot f(s,\boldsymbol{x},\boldsymbol{u},\boldsymbol{w}) + l(s,\boldsymbol{x},\boldsymbol{u},\boldsymbol{w})\}. \tag{2.3}$$

For pure target-reach problems, the running cost is set to zero and the target is represented by $\boldsymbol{g}$. The elapsed time $t = T - s$ converts the terminal condition into an initial condition. Write $\boldsymbol{v}(t,\boldsymbol{x}) = \boldsymbol{v}_{\mathrm{phys}}(T-t,\boldsymbol{x})$ and define

$$\boldsymbol{H}(t,\boldsymbol{x},\boldsymbol{p}) = -\boldsymbol{H}_{\mathrm{phys}}(T-t,\boldsymbol{x},\boldsymbol{p}). \tag{2.4}$$

This sign change is necessary: the physical equation is $\partial_s \boldsymbol{v}_{\mathrm{phys}} + \boldsymbol{H}_{\mathrm{phys}}(s,\boldsymbol{x},D\boldsymbol{v}_{\mathrm{phys}}) = 0$, while the elapsed equation is $\boldsymbol{v}_t + \boldsymbol{H}(t,\boldsymbol{x},D\boldsymbol{v}) = 0$. Fixing $\delta > 0$ gives the well-posed parabolic regularization

$$\boldsymbol{v}^\delta_t + \boldsymbol{H}(t,\boldsymbol{x},D\boldsymbol{v}^\delta) = \frac{\delta}{2}\Delta \boldsymbol{v}^\delta, \qquad \boldsymbol{v}^\delta(0,\boldsymbol{x}) = \boldsymbol{g}(\boldsymbol{x}). \tag{2.5}$$

Viscosity-solution theory supplies comparison and approximation results under additional hypotheses on the Hamiltonian and initial condition [10, 11]. The transformation below concerns classical positive-time solutions of this regularized equation; it does not prove their regularity from bounded initial conditions alone.

(a) HJ Linearization

Write $\boldsymbol{v} = \boldsymbol{v}^\delta$ within this subsection and choose a prescribed coefficient $\boldsymbol{c} \in C^{1,2}_b(\boldsymbol{x},t)$ with $0 < \boldsymbol{c}_- \le \boldsymbol{c} \le \boldsymbol{c}_+$. Define the transformed value and its initial condition by

$$\boldsymbol{\omega} = e^{-\boldsymbol{c}\boldsymbol{v}}, \qquad \boldsymbol{\omega}_0(\boldsymbol{x}) = e^{-\boldsymbol{c}(0,\boldsymbol{x})\boldsymbol{g}(\boldsymbol{x})}. \tag{2.6}$$

The coefficient need not match the Hamiltonian. A constant positive coefficient avoids coefficient derivatives and the singular ratio $\boldsymbol{H}/|D\boldsymbol{v}|^2$, while a variable coefficient permits local scaling. Neither choice removes the Hamiltonian's nonlinearity automatically. For a scalar $z > 0$ and vector $q \in \mathbb{R}^n$, introduce

$$V(t,\boldsymbol{x},z) = -\frac{\log z}{\boldsymbol{c}(t,\boldsymbol{x})}, \qquad P(t,\boldsymbol{x},z,q) = -\frac{q}{\boldsymbol{c}z} + \frac{\log z}{\boldsymbol{c}^2} D\boldsymbol{c}, \tag{2.7}$$

$$\boldsymbol{R}_{\mathrm{alg}}[\boldsymbol{c},V,P] = \boldsymbol{c}\left(\boldsymbol{H}(t,\boldsymbol{x},P) - \frac{\delta \boldsymbol{c}}{2}|P|^2\right), \tag{2.8}$$

$$\boldsymbol{R}_{\mathrm{der}}[\boldsymbol{c},V,P] = -\boldsymbol{c}_t V + \frac{\delta}{2}(\Delta \boldsymbol{c})V - \frac{\delta}{2}|D\boldsymbol{c}|^2 V^2 + \delta(1 - \boldsymbol{c}V) D\boldsymbol{c} \cdot P, \tag{2.9}$$

$$F(t,\boldsymbol{x},z,q) = z\{\boldsymbol{R}_{\mathrm{alg}}[\boldsymbol{c},V,P] + \boldsymbol{R}_{\mathrm{der}}[\boldsymbol{c},V,P]\}. \tag{2.10}$$

Algorithm 1 Full-residual source iteration on a short interval

1: Choose $\delta > 0$, a prescribed smooth coefficient $\boldsymbol{c} \geq \boldsymbol{c}_- > 0$, and an interval $[0, \tau]$.
2: Set $\boldsymbol{\omega}_0 = \exp(-\boldsymbol{c}(0,\cdot)\boldsymbol{g})$ and $z_0(t) = S_t\boldsymbol{\omega}_0$.
3: Specify a positive invariant admissible set containing z_0 and a contraction bound $\theta < 1$ for the full source map.
4: **for** $k = 0, 1, \ldots$ **do**
5: Evaluate $f_k = F(\cdot, z_k, Dz_k)$ on the required space–time domain.
6: Compute the value and gradient updates (2.13*a*)–(2.13*b*).
7: Bound sampling, quadrature, and reconstruction errors; verify admissibility and positivity.
8: Recover the value and gradient using (2.14).
9: Stop when the controlled error budget meets tolerance; otherwise continue or shorten the interval.
10: **end for**

The full transformed equation is

$$\boldsymbol{\omega}_t - \frac{\delta}{2}\Delta\boldsymbol{\omega} = F(t, \boldsymbol{x}, \boldsymbol{\omega}, D\boldsymbol{\omega}), \qquad \boldsymbol{\omega}(0) = \boldsymbol{\omega}_0. \tag{2.11}$$

For positive classical solutions, logarithmic inversion recovers the original viscous equation without discarding either residual. The heat equation follows only when their sum vanishes along the solution.

Let S_t denote the whole-space heat semigroup with covariance $\delta t I_n$. The transformed equation admits the mild representation,

$$\boldsymbol{\omega}(t) = S_t\boldsymbol{\omega}_0 + \int_0^t S_{t-s}F(s, \cdot, \boldsymbol{\omega}(s), D\boldsymbol{\omega}(s))\, ds, \tag{2.12}$$

coupling values and gradients at intermediate times. A solution of this semigroup integral equation is called a mild solution. Substituting the preceding iterate into its source defines one mild update. Thus, a single Gaussian average of the initial condition cannot replace this problem. Source iteration evaluates the right-hand side using the preceding space–time iterate, with coefficients fixed throughout each solve. Short-time contraction requires bounds on the source and its gradient dependence.

The source representation also gives the full spatial derivative. With $f_k(s, \boldsymbol{x}) = F(s, \boldsymbol{x}, z_k, Dz_k)$, one update is

$$z_{k+1}(t) = S_t\boldsymbol{\omega}_0 + \int_0^t S_{t-s}f_k(s)\, ds, \tag{2.13a}$$

$$Dz_{k+1}(t) = DS_t\boldsymbol{\omega}_0 + \int_0^t DS_{t-s}f_k(s)\, ds. \tag{2.13b}$$

$$\boldsymbol{v}_{k+1}(t, \boldsymbol{x}) = -\frac{\log z_{k+1}(t, \boldsymbol{x})}{\boldsymbol{c}(t, \boldsymbol{x})}, \qquad D\boldsymbol{v}_{k+1} = -\frac{Dz_{k+1}}{\boldsymbol{c}z_{k+1}} - \frac{\boldsymbol{v}_{k+1}}{\boldsymbol{c}}D\boldsymbol{c}. \tag{2.14}$$

The source integral and the derivative of the coefficient remain in these formulas. Appendix A derives their Gaussian forms. Bounded transformed initial conditions suffice for a bounded-source mild representation. Uniform logarithmic stability also requires a positive lower bound; an unbounded signed-distance initial condition does not supply that bound automatically.

The coefficient is prescribed throughout this iteration. Computing a coefficient from a previous iterate does not erase its derivatives. The one-dimensional implementation accompanying the new diagnostic uses exponential-Euler time stepping, rather than the whole-interval iteration in Algorithm 1. A production implementation still needs source and gradient values away from the displayed evaluation states.

(b) Relations to Backward Reachable Sets/Tubes

Let the target and backward reachable tube satisfy

$$\mathcal{L}_0 = \{\boldsymbol{x} : \boldsymbol{g}(\boldsymbol{x}) \le 0\}, \tag{2.15a}$$

$$\mathcal{L}([s,T],\mathcal{L}_0) = \{\boldsymbol{x} : \boldsymbol{v}_{\mathrm{tube,phys}}(s,\boldsymbol{x}) \le 0\}. \tag{2.15b}$$

For zero running cost, the tube value replaces the terminal payoff by $\min_{r\in[s,T]} \boldsymbol{g}(\xi(r))$. Under the game's reachability characterization, its zero sublevel set represents target entry during the interval [41, 42]. In particular,

$$\boldsymbol{x} \in \mathcal{L}([s,T],\mathcal{L}_0) \quad \Longleftrightarrow \quad \boldsymbol{v}_{\mathrm{tube,phys}}(s,\boldsymbol{x}) \le 0. \tag{2.16}$$

When the minimizing strategy and worst-case payoff are attained, the tube also has the strategy characterization

$$\mathcal{L}([s,T],\mathcal{L}_0) = \{\boldsymbol{x} : \exists \beta \in \mathcal{B}(s)\ \forall \boldsymbol{u} \in \overline{\mathcal{U}}(s)\ \exists r \in [s,T],\ \xi_{s,\boldsymbol{x}}^{\boldsymbol{u},\beta[\boldsymbol{u}]}(r) \in \mathcal{L}_0\}. \tag{2.17}$$

Without attainment, a zero value need not produce an optimizing strategy; the corresponding closure or approximate-strategy convention must be specified.

A reachable set at a fixed terminal horizon uses the terminal payoff instead. It must not be identified with the tube, which permits entry at an intermediate time.

In physical time, the tube equation and terminal condition are

$$\partial_s \boldsymbol{v}_{\mathrm{tube,phys}} + \min\{0, \boldsymbol{H}_{\mathrm{phys}}(s,\boldsymbol{x},D\boldsymbol{v}_{\mathrm{tube,phys}})\} = 0, \qquad \boldsymbol{v}_{\mathrm{tube,phys}}(T) = \boldsymbol{g}. \tag{2.18}$$

The corresponding elapsed-time viscous problem is

$$\partial_t \boldsymbol{v}^{\delta}_{\mathrm{tube}} + \max\{0, \boldsymbol{H}(t,\boldsymbol{x},D\boldsymbol{v}^{\delta}_{\mathrm{tube}})\} = \frac{\delta}{2}\Delta \boldsymbol{v}^{\delta}_{\mathrm{tube}}, \qquad \boldsymbol{v}^{\delta}_{\mathrm{tube}}(0) = \boldsymbol{g}. \tag{2.19}$$

Thus the full source must use the entire elapsed tube Hamiltonian $\max\{0, \boldsymbol{H}\}$. Applying the transformation to the unmodified terminal-game Hamiltonian solves a different problem.

Reach–avoid problems also require trajectories to satisfy a state constraint until target entry. If $h(\boldsymbol{x}) \le 0$ describes that constraint, then both h and the target function enter the stopping payoff and its terminal condition. Their value satisfies an obstacle problem. The Cauchy-source formulas apply in its continuation region but do not enforce obstacle complementarity or the stopping rule. No reach–avoid guarantee follows from the Cauchy analysis alone.

(c) Weighted Importance Sampling of Reachable Sets

Importance sampling changes the Gaussian proposal while preserving the integral through a density ratio. For a displacement $\vartheta \in \mathbb{R}^n$, variance $\sigma^2 > 0$, and sample $\boldsymbol{y} \sim \mathcal{N}(\boldsymbol{x}+\vartheta, \sigma^2 I_n)$, the original-to-proposal ratio is

$$w_\vartheta(\boldsymbol{y}) = \exp\left(\frac{|\vartheta|^2 - 2\vartheta\cdot(\boldsymbol{y}-\boldsymbol{x})}{2\sigma^2}\right). \tag{2.20}$$

Use $\sigma^2 = \delta t$ for the initial-condition integral and $\sigma^2 = \delta(t-s)$ for the source integral at time s. Correcting only terminal samples leaves the source contribution unaccounted for. Proposal changes require new moment bounds; a large effective sample size is a diagnostic rather than an error certificate.

The evaluation count M and sample count N play different roles. The former counts queried states; the latter counts independent random draws at each query and update. Source evaluation also requires intermediate-time values and gradients at sampled states. Their reconstruction and storage must enter the cost analysis, so a single endpoint sample array does not establish the cost of the full nonlinear solve.

The deterministic game computations integrate the complete transformed equation using the discretization specified in Section 3(b). The Gaussian formulas above describe source-update

integrals; implementing a nonlinear stochastic solve also requires reconstruction of intermediate-time values and gradients.

(d) Sampling Complexity and Convergence Guarantees

The sampling guarantee concerns one specified source update. Conditional on the preceding iterate, suppose $a_0 \le \boldsymbol{\omega}_0 \le b_0$ and $|f_k| \le B_F$. Uniform source-time sampling gives the unbiased transformed-value sample

$$A = \boldsymbol{\omega}_0(\boldsymbol{x} + \sqrt{\delta t} Z) + t f_k(tU, \boldsymbol{x} + \sqrt{\delta t(1-U)} Z), \qquad U \sim \mathcal{U}(0,1), \quad Z \sim \mathcal{N}(0, I_n). \tag{2.21}$$

The variables U and Z are independent. Each sample lies in an interval of width $D = b_0 - a_0 + 2tB_F$. The sample mean is unbiased for the transformed update; its logarithm and normalized gradient ratio need not be unbiased.

Theorem 2.1 (Finite-query concentration for a fixed source)**.** *Fix M queries, a maximum of K updates, and $\eta \in (0,1)$. Assume the preceding source bounds hold conditionally at every update with common sample-width bound D_*. If each batch contains N conditionally independent samples with the stated common conditional law, then all transformed-value sampling errors are at most*

$$h_N = D_* \sqrt{\frac{\log(2MK/\eta)}{2N}} \tag{2.22}$$

with probability at least $1 - \eta$.

Proof. Apply Hoeffding's inequality conditionally to each batch. Take expectations and a union bound over the MK fixed evaluations. Dependence between queries does not invalidate this bound. □

Corollary 2.1 (Sample count and logarithmic recovery)**.** *For transformed-value tolerance $\varepsilon_\omega > 0$, it suffices to choose*

$$N \ge \frac{D_*^2}{2\varepsilon_\omega^2} \log \frac{2MK}{\eta}. \tag{2.23}$$

If the target mild update is at least $a > 0$ and $h_N < a$, then its sampled logarithmic inverse differs by at most $h_N / [\mathbf{c}_-(a - h_N)]$ on the same event.

Proof. Solve the concentration bound for N. The event gives a sampled transformed value of at least $a - h_N$, so the mean value theorem for the logarithm gives the recovery bound. □

These statements do not give a continuum or gradient error bound. The endpoint-weighted source-time proposal used by the prototype has interval width $b_0 - a_0 + 4tB_F$ for its value sample; Appendix A derives the corresponding finite-second-moment gradient sample. Unbounded adaptive iteration counts require a summable allocation of failure probabilities.

Theorem 2.2 (Conditional convergence of the full source map)**.** *Let $X_\tau = C([0,\tau]; \mathrm{BUC}^1)$ with the sum of the uniform value and gradient norms. Suppose a nonempty closed set $\mathcal{A} \subset X_\tau$ is invariant under the mild map, its functions satisfy $a \le z \le b$ and $|Dz| \le Q$ with $a > 0$, and the bounded source is uniformly Lipschitz in (z, Dz) with constant L. If*

$$\theta = L\left(\tau + 2\sqrt{n\tau/\delta}\right) < 1, \tag{2.24}$$

then the full source map has a unique fixed point in $\mathcal{A}$ and its iterates starting in $\mathcal{A}$ converge geometrically in X_τ. If that fixed point is classical, its inverse solves the viscous Hamilton–Jacobi equation.

Appendix B proves the contraction and gives an invariant-ball criterion and a posteriori stopping bound. Invariance, regularity, and controlled gradient reconstruction are necessary hypotheses.

(e) Notes on Monte Carlo and Viscosity Approximation Errors

A residual of the original equation provides an error certificate that accounts for every transformation term. For a bounded smooth approximation U with bounded defect

$$r_U = U_t + \boldsymbol{H}(t, \boldsymbol{x}, DU) - \frac{\delta}{2}\Delta U, \tag{2.25}$$

comparison gives a value-error bound by the initial-condition error plus $\int_0^t \|r_U(s)\|_\infty ds$. If an independent vanishing-viscosity theorem also gives a bound $C_{\mathrm{vis}}\sqrt{\delta}$, the combined budget is

$$E(t) = \|U(0) - \boldsymbol{g}\|_\infty + \int_0^t \|r_U(s)\|_\infty ds + C_{\mathrm{vis}}\sqrt{\delta}. \tag{2.26}$$

The approximation rate depends on the problem's hypotheses and constant; the number $\sqrt{\delta}$ alone is not a certificate. The transformed defect satisfies $d = -\boldsymbol{c}\exp(-\boldsymbol{c}U)r_U$, as proved in Appendix B. Sampled residuals require additional bounds between states and through time before they can supply the uniform norm in this budget.

Corollary 2.2 (Conditional sign certification)**.** *Suppose $|U - \boldsymbol{v}^0| \le E$ on a specified set, with $\boldsymbol{v}^0 \le 0$ denoting the target-reaching class. Then $U > E$ implies $\boldsymbol{v}^0 > 0$, and $U < -E$ implies $\boldsymbol{v}^0 < 0$. The remaining states form the abstention band $|U| \le E$.*

Proof. Subtract or add E to the assumed error bound. Each strict inequality then fixes the sign of $\boldsymbol{v}^0$. □

A probabilistic guarantee has the probability of the error event actually established. Finite-query concentration does not certify every state of a continuous reachable set. A reach–avoid interpretation also requires its obstacle problem.

Reducing δ can reduce viscosity bias while increasing source sensitivity, score variance, and logarithmic instability. The full-source Lipschitz and positivity constants also affect the admissible time interval. Neither a universal optimal power law for $\delta(N)$ nor a dimension-independent computational bound follows from these results. The global complexity theory for multilevel Picard methods has additional source and derivative-growth assumptions [23, 27]; Appendix B explains why they do not automatically hold here.

3. Numerical results

The experiments address two distinct questions: whether a discretization of the complete transformed equation resolves its stated multidimensional Cauchy problem, and how the resulting field can be evaluated at a large population of states. All game comparisons use the same viscosity, Hamiltonian, bounded initial condition, and horizon in both solvers. This separates numerical discrepancy from vanishing-viscosity bias. The one-dimensional manufactured verification is reported in Appendix C(e).

(a) Games of two vehicles on a plane

The pursuit–evasion examples use relative Dubins dynamics [38, 41] and a normalized rockets benchmark motivated by [16, 44]. Appendix C specifies the physical control optimization and its elapsed-time sign. Figure 1 compares the rockets value fields at three headings. The relative-Dubins example in Figure 3 provides a second game Hamiltonian. A single-Dubins pose target and a double-integrator target complete the control examples.

Table 1: Maximum sampled discrepancies for the full residual computations. Refinement and independent-reference columns use the common observation queries; the defect column gives the largest sampled original-equation defect over native states at three interior times. All payoffs are dimensionless. The grid column lists the fine spatial and heading counts. Pending or failed checks are disclosed in the reproducibility record.

System	Grid	Refinement	Reference	Defect	Status
Rockets	$160^2 \times 80$	0.007481	0.003223	0.1433	Pending
Relative Dubins	pending	pending	pending	pending	Pending
Single Dubins	$128^2 \times 64$	0.009297	0.001415	0.02992	Pass
Double integrator	128^2	0.006741	5.60×10^{-4}	0.01187	Pass
Bird model	pending	pending	pending	pending	Pending

The rows in Table 1 concern terminal-horizon Cauchy problems. A zero contour represents the numerical terminal reachable set for the stated positive-viscosity model. Intermediate-time target entry instead requires the stopping formulation in Section 2. The normalized rockets system is defined directly by its controlled vector field; the physical-unit illustrations and launch problem are given in [44].

(b) Refinement and sensitivity

The transformed solver integrates the complete source in (2.10) by second-order exponential Runge–Kutta time stepping on a periodic Fourier grid. A fixed cutoff retains source modes with component index at most one third of the corresponding grid count. This is a collocation approximation to the full source; the equation defect is evaluated against the untruncated source. A separate fourth-order finite-difference discretization with classical fourth-order Runge–Kutta time stepping evolves the original viscous equation. Positivity is checked at every transformed stage, without clipping.

All coordinates and payoffs in these normalized computations are dimensionless. For a circular target of radius r, the initial condition in the observation region is

$$\boldsymbol{g}(\boldsymbol{x}) = \tanh\left(\sqrt{\boldsymbol{x}_1^2 + \boldsymbol{x}_2^2 + \epsilon_g^2} - \sqrt{r^2 + \epsilon_g^2}\right), \qquad \epsilon_g = 0.08. \tag{3.1}$$

The single-Dubins pose target adds $4\sin^2(\theta/2)$ under the first square root. The bird example takes the minimum of seven translated radial payoffs, with target centers on a radius-1.15 ring and disk radius 0.6. This specifies the actual viscous initial condition; preserving a signed-distance function's initial zero set does not preserve its positive-time viscous solution.

The baseline spatial box is $[-6, 6]^2$, with periodic heading $[-\pi, \pi)$ for three-dimensional cases. A smooth completion makes the initial condition equal to 1 near each spatial seam while leaving it unchanged on $[-4.5, 4.5]^2$. The observation box is $[-2, 2]^2$ for the single-Dubins, relative-Dubins, and double-integrator cases and $[-3, 3]^2$ for rockets and birds. The prescribed coefficient is

$$\boldsymbol{c}(t, \boldsymbol{x}) = b + 0.04\cos t \prod_{j=1}^{d} \cos(2\pi \boldsymbol{x}_j / L_j), \qquad b = 0.2, \tag{3.2}$$

where L_j is the corresponding periodic length. The computation includes its time derivative, gradient, and Laplacian. The sensitivity calculation changes b to 0.35 while keeping the Hamiltonian and initial condition fixed.

Successive field refinements can change both spatial resolution and time step; fixed-grid time halving is checked separately. Refinement and solver discrepancies are measured on the same 65×65 observation points and, in three dimensions, 32 equally spaced headings. These are maximum sampled discrepancies of interpolated fields. Equation defects use native collocation

Table 2: Maximum sampled changes under time halving, coefficient variation, and combined box sensitivity. The last experiment changes the spatial box at fixed spacing and also changes the time step and coefficient wavelength. All quantities use the common observation queries and dimensionless payoffs.

System	Time	Coefficient	Combined box
Rockets	7.19×10^{-6}	1.84×10^{-4}	pending
Relative Dubins	pending	pending	pending
Single Dubins	1.47×10^{-6}	1.02×10^{-5}	8.83×10^{-6}
Double integrator	6.85×10^{-7}	1.73×10^{-6}	4.40×10^{-5}
Bird model	pending	pending	pending

states and centered differences at one quarter, one half, and three quarters of the elapsed horizon. They are finite diagnostics rather than uniform space–time bounds.

The acceptance thresholds were fixed before the refinement cycle: 0.01 for successive field refinements, 0.02 for the independent reference, 0.002 for time halving and coefficient changes, and 0.005 for the combined box-sensitivity experiment. Sampled original-equation defects must have maximum at most 0.15 and root-mean-square at most 0.02. The box experiment extends the half-width to 7.5 at fixed spatial spacing and heading resolution; it also changes the time step and coefficient wavelength, so it measures their combined sensitivity. Diffusion has infinite propagation speed, and this experiment is not an isolated boundary-error estimate.

(c) Population-scale evaluation of starling-inspired dynamics

The attacked-agent model in Appendix C(d) has four state components: two relative positions, altitude, and heading. With equal vertical control bounds and an altitude-independent payoff, the vertical Hamiltonian terms cancel and the solution is independent of altitude. The reduced three-dimensional Hamiltonian is therefore the relative-Dubins Hamiltonian. This identity is checked independently by enumerating the four-dimensional physical controls.

We evaluate the resulting field at pending freshly generated four-component states, using a fixed saved random seed. Positions are independently uniform on $[-3, 3]^2$, altitude on $[-1, 1]$, and heading on $[-\pi, \pi)$. Every queried value is retained, together with its independent-reference value. The population contains pending finite paired evaluations. Of these, pending have a positive approximate value; pending change sign between the two numerical fields. These counts describe the saved query population. They are not a simulation of coupled flock trajectories or a population-wide safety certificate. The corresponding geometry is shown in Appendix C(d).

(d) Computational scope

The experiments distinguish population size from joint state dimension. A model of 15 individually steered planar vehicles has the dynamics

$$\dot{x}_i = a_i \cos\theta_i, \quad \dot{y}_i = a_i \sin\theta_i, \quad \dot{\theta}_i = u_i, \qquad i = 1, \ldots, 15, \tag{3.3}$$

and a joint state in $\mathbb{R}^{45}$. A tensor grid with 101 points per coordinate would contain 101^{45} states. The present full residual computations resolve two- and three-dimensional fields, including an exact reduction of the four-component bird model. Evaluating one such field at many states does not establish accuracy or computational scaling for a joint 45-dimensional equation. Source reconstruction for general high-dimensional problems remains a separate computational requirement.

4. Conclusion

Retaining the transformation source preserves the viscous Hamilton–Jacobi equation under positive classical logarithmic inversion. The resulting mild representation includes both the initial-condition convolution and a nonlinear source integral. Its gradient also includes the coefficient derivative. Homogeneous heat flow remains a special case, rather than the transformed equation for every game Hamiltonian.

The full source iteration converges on a short interval under explicit invariance, positivity, and value-and-gradient Lipschitz hypotheses. Finite-query sampling bounds control specified source evaluations, while a uniform original-equation residual supplies a conditional sign certificate. Neither result establishes a continuum guarantee from sampled errors alone.

The numerical verification separates manufactured-solution accuracy, agreement between independent discretizations, field refinement, and sensitivity to time step, coefficient, and periodic-box choices. The direct transformed solver retains every analytic residual term. Its finite-dimensional source cutoff is checked against the untruncated equation defect. The bird example reduces exactly from four state components to three under the stated altitude-invariance assumptions, allowing evaluation at many saved population states. General high-dimensional source reconstruction and uniform population safety require additional error control.

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Appendices

Contents

A Transformation and sampling formulas **15**
(a) Sampling a fixed source 16
(b) Linear potential solves 16
B Convergence, residual certification, and robustness **18**
(a) Convergence of source iteration 18
(b) Sampling and accumulated implementation error 19
(c) A posteriori certificate for the original equation 20
(d) Viscosity, model perturbations, and level sets 20
C Further Numerical Results **22**
(a) Relative Dubins pursuit–evasion 22
(b) The double integrator 24
Dynamics and Problem Setup 24
Time-optimal control scheme 25
(c) A normalized rockets benchmark 25
(d) Population-scale evaluation for starling-inspired dynamics 26
Notations and Background 27
Murmuration as a Levelsets Fronts Equation 28
Related Works 28
Problem Formulation 29
Framework for Separated Payoffs. 30
Global Isotropy via Local Anisotropy. 30
Flock Motion from Aggregated Value Functions. 31
Vacuole nucleation diagnostics 32
Flock splitting diagnostics 32
Cordon geometry 32
Flash expansion diagnostics 32
Markers in numerical slices 33
Numerical geometry and its interpretation 33
(e) Validation of the retained transformation source 33
Multidimensional manufactured verification 33
D Limitations **35**

A. Transformation and sampling formulas

The full residual changes the representation formula rather than the chain rule. Throughout this appendix, the domain is $\mathbb{R}^n$, the diffusion coefficient is $\delta/2$, and $\boldsymbol{c}$ satisfies the bounds in §2(a). Bounded-domain problems require the corresponding boundary semigroup and boundary data.

Lemma A.1 (Transformation identity)**.** *Suppose $\boldsymbol{v}, \boldsymbol{c} \in C^{1,2}$ and set $\boldsymbol{\omega} = e^{-\boldsymbol{c}\boldsymbol{v}}$. Define the original-equation defect $r_v = \boldsymbol{v}_t + \boldsymbol{H}(t, \boldsymbol{x}, D\boldsymbol{v}) - \delta\Delta\boldsymbol{v}/2$. Then*

$$\boldsymbol{\omega}_t - \frac{\delta}{2}\Delta\boldsymbol{\omega} = \boldsymbol{\omega}\{\boldsymbol{R}_{\mathrm{alg}}[\boldsymbol{c}, \boldsymbol{v}, D\boldsymbol{v}] + \boldsymbol{R}_{\mathrm{der}}[\boldsymbol{c}, \boldsymbol{v}, D\boldsymbol{v}] - \boldsymbol{c}r_v\}. \tag{A.1}$$

Consequently, positive classical solutions of the full transformed equation and classical solutions of the original equation correspond under logarithmic inversion.

Proof. Set $\phi = \boldsymbol{c}\boldsymbol{v}$. Differentiation gives $\boldsymbol{\omega}_t = -\boldsymbol{\omega}\phi_t$, $D\boldsymbol{\omega} = -\boldsymbol{\omega}D\phi$, and $\Delta\boldsymbol{\omega} = \boldsymbol{\omega}(|D\phi|^2 - \Delta\phi)$. Expand $D\phi = \boldsymbol{c}D\boldsymbol{v} + \boldsymbol{v}D\boldsymbol{c}$ and $\Delta\phi = \boldsymbol{c}\Delta\boldsymbol{v} + 2D\boldsymbol{c}\cdot D\boldsymbol{v} + \boldsymbol{v}\Delta\boldsymbol{c}$. Substitution cancels the two terms containing $\boldsymbol{c}\Delta\boldsymbol{v}$. The remaining terms give (A.1). Since $\boldsymbol{c}\boldsymbol{\omega} > 0$, the original defect vanishes if and only if the transformed defect vanishes. □

Corollary A.1 (Reduction to homogeneous heat flow)**.** *For a solution of the original equation, homogeneous heat flow holds if and only if $\boldsymbol{R}_{\mathrm{alg}} + \boldsymbol{R}_{\mathrm{der}} = 0$ along that solution. A sufficient condition is a constant coefficient $\boldsymbol{c}_0 > 0$ and $\boldsymbol{H}(t, \boldsymbol{x}, D\boldsymbol{v}) = \delta\boldsymbol{c}_0|D\boldsymbol{v}|^2/2$ along the solution. The Hamiltonian $\boldsymbol{H}(p) = |p|^2/2$ satisfies this condition with $\boldsymbol{c}_0 = 1/\delta$.*

Proof. Apply Lemma A.1 with $r_v = 0$. Constancy of $\boldsymbol{c}$ removes the derivative residual, and the stated quadratic relation removes the algebraic residual. □

A constant $\boldsymbol{c}$ is not necessary for a particular solution. For example, $\boldsymbol{v} = 0$ and $\boldsymbol{H}(t, \boldsymbol{x}, 0) = 0$ give $\boldsymbol{\omega} = 1$ and zero total residual for every admissible variable $\boldsymbol{c}$. Requiring both residuals to vanish separately also excludes possible cancellation between them.

Corollary A.2 (Source representation and value recovery)**.** *Suppose $\boldsymbol{\omega}_0$ is bounded and uniformly continuous, and the source $f(s, \boldsymbol{x}) = F(s, \boldsymbol{x}, \boldsymbol{\omega}(s, \boldsymbol{x}), D\boldsymbol{\omega}(s, \boldsymbol{x}))$ is bounded and continuous. Every bounded classical solution satisfies*

$$\boldsymbol{\omega}(t, \boldsymbol{x}) = \mathbb{E}[\boldsymbol{\omega}_0(\boldsymbol{x} + \sqrt{\delta t}\, Z)] + \int_0^t \mathbb{E}[f(s, \boldsymbol{x} + \sqrt{\delta(t-s)}\, Z)]\, ds, \tag{A.2a}$$

$$\boldsymbol{v}(t, \boldsymbol{x}) = -\frac{1}{\boldsymbol{c}(t, \boldsymbol{x})}\log \boldsymbol{\omega}(t, \boldsymbol{x}), \qquad Z \sim \mathcal{N}(0, I_n). \tag{A.2b}$$

Proof. Variation of constants for the heat equation gives the mild representation. The whole-space heat kernel has unit mass and equals the indicated Gaussian density. Invert the transformation at the evaluation point. □

The source contains the unknown solution. If $|\boldsymbol{v}| \le G$, then $e^{-\boldsymbol{c}_+ G} \le \boldsymbol{\omega} \le e^{\boldsymbol{c}_+ G}$. A lower bound on an unbounded signed-distance initial condition alone does not supply this uniform lower bound.

Proposition A.1 (Gradient with the full source)**.** *Under the preceding bounded-source hypotheses, for $t > 0$ the gradient satisfies*

$$D\boldsymbol{\omega}(t, \boldsymbol{x}) = \frac{1}{\sqrt{\delta t}}\mathbb{E}[Z\boldsymbol{\omega}_0(\boldsymbol{x} + \sqrt{\delta t}\, Z)] + \int_0^t \frac{\mathbb{E}[Z f(s, \boldsymbol{x} + \sqrt{\delta(t-s)}\, Z)]}{\sqrt{\delta(t-s)}}\, ds, \tag{A.3a}$$

$$D\boldsymbol{v}(t, \boldsymbol{x}) = -\frac{D\boldsymbol{\omega}(t, \boldsymbol{x})}{\boldsymbol{c}(t, \boldsymbol{x})\boldsymbol{\omega}(t, \boldsymbol{x})} - \frac{\boldsymbol{v}(t, \boldsymbol{x})}{\boldsymbol{c}(t, \boldsymbol{x})}D\boldsymbol{c}(t, \boldsymbol{x}). \tag{A.3b}$$

Proof. Differentiate the Gaussian kernel with respect to its center. Its derivative has integrable L^1 norm bounded by $\sqrt{n/(\delta(t-s))}$. This bound justifies differentiation of the source integral. The product rule applied to $\boldsymbol{v} = -\boldsymbol{c}^{-1}\log\boldsymbol{\omega}$ gives the second formula. □

(a) Sampling a fixed source

Monte Carlo sampling estimates a specified source integral. It does not evaluate the unknown source without an approximation of the space–time solution. Conditional on a fixed preceding iterate, write f for its deterministic source. For independent $U_i \sim \mathcal{U}(0,1)$ and $Z_i \sim \mathcal{N}(0, I_n)$, define

$$A_i = \boldsymbol{\omega}_0(\boldsymbol{x} + \sqrt{\delta t}Z_i) + tf(tU_i, \boldsymbol{x} + \sqrt{\delta t(1-U_i)}Z_i), \qquad \widehat{\boldsymbol{\omega}}_N = \frac{1}{N}\sum_{i=1}^{N} A_i. \tag{A.4}$$

The mean is unbiased for one mild update. The two terms may share the Gaussian draw, since independence is needed between the sample pairs, rather than within each pair. Neither $-\boldsymbol{c}^{-1}\log\widehat{\boldsymbol{\omega}}_N$ nor a ratio gradient estimator is generally unbiased. The logarithm requires a positive estimate.

Uniform source-time sampling gives a score gradient whose second moment need not be finite near $s = t$. Instead, draw the time from

$$\pi_t(s) = \frac{1}{2\sqrt{t}\sqrt{t-s}}, \quad 0 < s < t, \qquad S = t(1-U^2). \tag{A.5}$$

Using independent copies of S, Z, a gradient sample is

$$B = \frac{Z_0\boldsymbol{\omega}_0(\boldsymbol{x} + \sqrt{\delta t}Z_0)}{\sqrt{\delta t}} + 2\sqrt{\frac{t}{\delta}}\, Z_1 f(S, \boldsymbol{x} + \sqrt{\delta(t-S)}Z_1). \tag{A.6}$$

Its mean equals (A.3*a*). If $|\boldsymbol{\omega}_0| \le W$ and $|f| \le B_f$, then its second moment obeys

$$\mathbb{E}|B|^2 \le \frac{2nW^2}{\delta t} + \frac{8ntB_f^2}{\delta}. \tag{A.7}$$

The bound follows from $|a+b|^2 \le 2|a|^2 + 2|b|^2$ and $\mathbb{E}|Z|^2 = n$. No bounded-gradient concentration claim follows from a bounded value sample alone.

Endpoint importance sampling remains valid only after including the density ratio for each sampled integral. For example, for a Gaussian displacement h and variance σ^2, the original-to-shifted density ratio is

$$\frac{\varphi_{\boldsymbol{x}}(y)}{\varphi_{\boldsymbol{x}+h}(y)} = \exp\left(\frac{|h|^2 - 2h\cdot(y-\boldsymbol{x})}{2\sigma^2}\right). \tag{A.8}$$

Apply this ratio at each source time with $\sigma^2 = \delta(t-s)$. A tilt of terminal samples alone does not correct the source integral. Importance sampling may change the variance and requires its own moment bounds.

(b) Linear potential solves

A prescribed bounded continuous potential $\boldsymbol{R}$ and bounded initial condition give another representation. For the bounded mild solution, with $X_r = \boldsymbol{x} + \sqrt{\delta}W_r$, the solution of $\boldsymbol{\omega}_t = \delta\Delta\boldsymbol{\omega}/2 + \boldsymbol{R}\boldsymbol{\omega}$ satisfies

$$\boldsymbol{\omega}(t,\boldsymbol{x}) = \mathbb{E}_{\boldsymbol{x}}\left[\boldsymbol{\omega}_0(X_t)\exp\left\{\int_0^t \boldsymbol{R}(t-r, X_r)\,dr\right\}\right]. \tag{A.9}$$

Freezing $\boldsymbol{R}[\boldsymbol{c}, \boldsymbol{v}^{(k)}, D\boldsymbol{v}^{(k)}]$ produces a linear potential equation, whose positive solution defines a possible iteration. For the full equation, however, $\boldsymbol{R}$ depends on the unknown value and gradient. The formula then remains implicit. Its potential iteration is distinct from the additive source iteration analyzed below, and requires a separate convergence proof. A stable log-sum-exp

evaluation applies to positive path weights after time discretization; it does not remove the path quadrature error.

B. Convergence, residual certification, and robustness

Retaining the source removes the defect caused by deliberately dropping transformation terms. It does not remove errors from iteration, sampling, differentiation, or time and space approximation. This appendix gives conditional convergence and a residual certificate for the equation that the revised method solves.

(a) Convergence of source iteration

Gradient dependence requires a norm that controls both values and first derivatives. Let $\mathrm{BUC}^1(\mathbb{R}^n)$ contain continuously differentiable functions with bounded, uniformly continuous values and gradients. On $X_\tau = C([0,\tau];\mathrm{BUC}^1)$, use

$$\|z\|_{X_\tau} = \sup_{s\le\tau}\|z(s)\|_\infty + \sup_{s\le\tau}\|Dz(s)\|_\infty. \tag{B.1}$$

For $\boldsymbol{g}\in\mathrm{BUC}^1$ and bounded coefficient derivatives, the transformed initial condition belongs to the same space. Define the mild update by

$$(\mathcal{M}z)(t) = S_t\boldsymbol{\omega}_0 + \int_0^t S_{t-s}F(s,\cdot,z(s),Dz(s))\,ds. \tag{B.2}$$

Theorem B.1 (Conditional short-time convergence)**.** *Let $\mathcal{A}\subset X_\tau$ be nonempty, closed, and invariant under $\mathcal{M}$. Suppose all its members satisfy $a\le z\le b$ and $|Dz|\le Q$, with $a>0$. Assume that the source is bounded and uniformly continuous on the relevant coefficient and state ranges, and satisfies*

$$|F(t,\boldsymbol{x},z,q) - F(t,\boldsymbol{x},\widetilde{z},\widetilde{q})| \le L\big(|z-\widetilde{z}| + |q-\widetilde{q}|\big). \tag{B.3}$$

If $\theta = L(\tau + 2\sqrt{n\tau/\delta}) < 1$, then $\mathcal{M}$ has a unique fixed point in $\mathcal{A}$. Its iterates satisfy

$$\|z^{(K)}-\boldsymbol{\omega}\|_{X_\tau} \le \theta^K\|z^{(0)}-\boldsymbol{\omega}\|_{X_\tau}, \qquad \|z^{(K)}-\boldsymbol{\omega}\|_{X_\tau} \le \frac{\theta}{1-\theta}\|z^{(K)}-z^{(K-1)}\|_{X_\tau} \tag{B.4}$$

for $K\ge 1$. If the fixed point is classical, its inverse solves the original viscous equation.

Proof. The heat semigroup contracts the supremum norm. Gaussian differentiation and $\mathbb{E}|Z|\le\sqrt{n}$ give

$$\|DS_r h\|_\infty \le \sqrt{\frac{n}{\delta r}}\|h\|_\infty. \tag{B.5}$$

Apply the source Lipschitz bound inside the difference of two mild updates. The value term contributes $L\tau$ and the gradient term contributes $2L\sqrt{n\tau/\delta}$. Their sum proves contraction in X_τ. Completeness and invariance permit Banach's theorem. Summing successive differences gives the posterior bound. Lemma A.1 identifies every positive classical fixed point with the original equation. □

For bounded coefficient derivatives, $\boldsymbol{c}\ge\boldsymbol{c}_-> 0$, and $z\ge a>0$, the inverse maps have bounded derivatives on bounded (z,q) ranges. A Hamiltonian uniformly bounded and locally Lipschitz in p on those ranges, uniformly over $(t,\boldsymbol{x})$, therefore supplies a finite source constant L. No convexity assumption is needed for this contraction argument.

The global complexity theorem in [27, Corollary 4.8] does not follow from this local argument. Their setting imposes a globally Lipschitz source in the value and gradient variables, together with strong smoothness and derivative-growth bounds for the complexity result. The transformed source need not meet those assumptions. Even with constant $\boldsymbol{c}$ and the linear Hamiltonian $\boldsymbol{H}(p) = ap$ in one dimension, it becomes

$$F(z,q) = -aq - \frac{\delta}{2z}q^2. \tag{B.6}$$

This source is singular as z approaches zero and is quadratic in q. Using a globally Lipschitz extension to analyze the original equation would require a proof that the solution remains inside

the bounded positive region on which the sources agree. Transferring their complexity result would also require verifying their dimension-dependent constants and quadrature regularity. The present contraction theorem establishes neither requirement.

Invariance is a substantive hypothesis. One sufficient construction starts with $b_0(t) = S_t\boldsymbol{\omega}_0$ and an X_τ ball of radius r about b_0. Choose $r < \inf \boldsymbol{\omega}_0$ to preserve positivity. If $|F| \le B_F$ throughout that ball and

$$B_F\left(\tau + 2\sqrt{n\tau/\delta}\right) \le r, \tag{B.7}$$

then the mild map preserves the ball. Under the preceding uniform-in-state Hamiltonian and coefficient assumptions, bounded initial values and gradients give finite source ranges for sufficiently small τ. The case of a constant $\boldsymbol{c} = 1/\delta$ and $\boldsymbol{H} = |p|^2/2$ has $F = 0$ and $\theta = 0$, so the hypotheses include a nonempty solvable case. Extension to a longer horizon requires renewed bounds and invariance on each interval; a global contraction is not asserted.

If $0 < \theta < 1$ and the initial error is bounded by $E_0 > 0$, a sufficient iteration count is

$$K \ge \max\left\{0, \left\lceil \frac{\log(E_0/\varepsilon)}{\log(1/\theta)} \right\rceil \right\}. \tag{B.8}$$

When $\theta = 0$, one update reaches the fixed point. A small change between noisy iterates alone does not certify convergence.

(b) Sampling and accumulated implementation error

For the fixed-source value samples in (A.4), suppose $a_0 \le \boldsymbol{\omega}_0 \le b_0$ and $|f| \le B_f$. Each sample lies in an interval of width $D = b_0 - a_0 + 2tB_f$. At M fixed queries, Hoeffding's inequality and a union bound give

$$\max_{1\le m\le M} |\widehat{\boldsymbol{\omega}}_N(t_m, \boldsymbol{x}_m) - (\mathcal{M}z)(t_m, \boldsymbol{x}_m)| \le D_*\sqrt{\frac{\log(2M/\eta)}{2N}} \tag{B.9}$$

with probability at least $1 - \eta$, using a common width bound D_*. Independence between different queries is unnecessary. The preceding iterate must be fixed conditional on the past, and the new samples must have the stated conditional law. A union bound over K adaptive iterations replaces M by MK.

The prototype instead uses the source-time density π_t for both value and gradient samples. Its value-source weight is $1/\pi_t(S) \le 2t$, so the corresponding interval width is $b_0 - a_0 + 4tB_f$. The same concentration argument applies with that larger width.

This is a finite-query value bound. A continuum X_τ guarantee also requires gradient concentration and control of reconstruction between queries. Gaussian score samples are unbounded, so the same Hoeffding argument does not establish that guarantee. The second-moment bound in (A.7) supports variance estimates, but does not by itself control a continuum supremum.

If a numerical implementation separately establishes X_τ errors ϵ_k in each mild update and keeps its iterates in the admissible set, then contraction gives

$$\|\widehat{z}^{(K)} - \boldsymbol{\omega}\|_{X_\tau} \le \theta^K E_0 + \sum_{j=0}^{K-1} \theta^{K-1-j}\epsilon_j. \tag{B.10}$$

The proof iterates the inequality $e_{k+1} \le \theta e_k + \epsilon_k$. Thus sampling, source quadrature, and reconstruction errors accumulate through the solve. They do not enter only as a final independent sample average. If both transformed values are at least a, logarithmic inversion yields

$$\|\widehat{\boldsymbol{v}}^{(K)} - \boldsymbol{v}^\delta\|_\infty \le \frac{\theta^K E_0 + \sum_{j=0}^{K-1} \theta^{K-1-j}\epsilon_j}{\boldsymbol{c} - a}. \tag{B.11}$$

This conditional bound does not certify the accompanying numerical prototype, which uses time stepping rather than whole-interval Picard iteration.

(c) A posteriori certificate for the original equation

A residual of the full transformed equation measures the defect of the original equation directly. Assume comparison for the original parabolic equation on the whole space. Let U be a bounded classical approximation with bounded residual, and define

$$w = e^{-\boldsymbol{c}U}, \qquad d = w_t - \frac{\delta}{2}\Delta w - F(t, \boldsymbol{x}, w, Dw). \tag{B.12}$$

Theorem B.2 (Residual certificate)**.** *Under these hypotheses, the viscous solution with initial condition* $\boldsymbol{g}$ *obeys*

$$\|U(t) - \boldsymbol{v}^{\delta}(t)\|_{\infty} \le \|U(0) - \boldsymbol{g}\|_{\infty} + \int_0^t \left\| \frac{d(s)}{\boldsymbol{c}(s)w(s)} \right\|_{\infty} ds. \tag{B.13}$$

Proof. Lemma A.1 gives $d = -\boldsymbol{c}wr_U$, with $r_U = U_t + \boldsymbol{H}(t, \boldsymbol{x}, DU) - \delta\Delta U/2$. Let $e_0 = \|U(0) - \boldsymbol{g}\|_{\infty}$ and $E(t) = e_0 + \int_0^t \|r_U(s)\|_{\infty} ds$. Since the Hamiltonian does not depend on the value, $U - E$ is a subsolution and $U + E$ is a supersolution. Their initial condition bracket $\boldsymbol{g}$. Comparison gives $U - E \le \boldsymbol{v}^{\delta} \le U + E$, proving the bound. □

The source must include both residual components when computing d. If a solver omits $\boldsymbol{R}_{\mathrm{der}}$, then its transformed defect contains the missing term $-w\boldsymbol{R}_{\mathrm{der}}$. The certificate charges that omission. Residuals evaluated only at sampled states are diagnostic; rigorous use requires a bound between those states and through time, including derivative approximation error.

(d) Viscosity, model perturbations, and level sets

Vanishing viscosity remains a separate approximation. Suppose the Hamiltonian and initial condition satisfy a comparison and approximation theorem giving $\|\boldsymbol{v}^{\delta} - \boldsymbol{v}^{0}\|_{\infty} \le C_{\mathrm{vis}}\sqrt{\delta}$ on the chosen horizon. This rate is an additional hypothesis, rather than a consequence of the transformation. Under the same uniform norm and domain, Theorem B.2 then gives

$$\|U(t) - \boldsymbol{v}^{0}(t)\|_{\infty} \le \|U(0) - \boldsymbol{g}\|_{\infty} + \int_0^t \left\| \frac{d(s)}{\boldsymbol{c}(s)w(s)} \right\|_{\infty} ds + C_{\mathrm{vis}}\sqrt{\delta}. \tag{B.14}$$

No universal choice $\delta \sim N^{-1/3}$ follows: source, stability, and sampling constants can depend on δ. Their dependence must be established before optimizing a combined rate.

Proposition B.1 (Hamiltonian and initial condition perturbations)**.** *Suppose two equations with the same diffusion admit comparison, their initial conditions differ by at most* ϵ_g*, and* $|\boldsymbol{H}_1(t, \boldsymbol{x}, p) - \boldsymbol{H}_2(t, \boldsymbol{x}, p)| \le \epsilon_H$ *throughout the required co-state range. For classical solutions, it suffices that this range contains* $D\boldsymbol{v}_2$*. Then*

$$\|\boldsymbol{v}_1(t) - \boldsymbol{v}_2(t)\|_{\infty} \le \epsilon_g + t\epsilon_H. \tag{B.15}$$

Proof. The functions $\boldsymbol{v}_2 - \epsilon_g - t\epsilon_H$ and $\boldsymbol{v}_2 + \epsilon_g + t\epsilon_H$ are a subsolution and a supersolution for the first equation. Comparison proves the claim. For viscosity solutions, impose the Hamiltonian bound on the test-gradient range required by comparison, or uniformly over all p. □

Changing a prescribed positive transformation coefficient does not perturb the Hamiltonian when the full source is retained. Thus a smooth positive regularization that preserves the stated differentiability changes coordinates; it creates no Hamiltonian modeling error by itself. Hard clipping can destroy that differentiability and falls outside these classical results. If the Hamiltonian is deliberately replaced, the perturbation bound applies to that separate change. The matching choice $\boldsymbol{c} = 2\boldsymbol{H}(t, \boldsymbol{x}, D\boldsymbol{v})/(\delta|D\boldsymbol{v}|^2)$ can vanish, change sign, or become singular. At $\boldsymbol{c} = 0$, the exponential transform loses the value information. Recomputing this ratio also requires regularity beyond the prescribed-coefficient hypotheses. The full source permits a positive constant coefficient even in these cases.

If a uniform bound $|U - \boldsymbol{v}^0| \leq E$ has been established on a specified domain, then

$$\{U < -E\} \subseteq \{\boldsymbol{v}^0 < 0\}, \qquad \{U > E\} \subseteq \{\boldsymbol{v}^0 > 0\}. \tag{B.16}$$

These inclusions follow by subtraction of the error bound. A finite-query sampling event supplies the inclusions only at those queries. The uncertain band $|U| \leq E$ requires abstention. Interpreting either sign as safety also requires the corresponding game's value convention and, for reach–avoid problems, a correctly solved obstacle problem.

C. Further Numerical Results

The model descriptions connect the game experiments to their dynamical and geometric interpretation. All recomputed fields use the full residual and the elapsed-time convention of Section 2. Numerical discrepancies and sampled equation defects are distinguished from uniform value-error certificates.

(a) Relative Dubins pursuit–evasion

This example was originally proposed by [38] as an extension of the homicidal chauffeur game of Isaacs [29], whereupon a pursuit-evasion game between two players with similar speeds and minimum turn radii, is thoroughly analyzed. In [41], this problem was established as a benchmark for testing the solubility of capturable set of states (the backward reachable tube) in Merz's classical pursuit-evasion game. These results supply the game context. The present computation evaluates positive-viscosity terminal-horizon values using the full residual equation.

The game is that of two cars sharing similar Dubins dynamics [17]: $\boldsymbol{P}$ and $\boldsymbol{E}$ both have a positive minimum turn radii, w, and constant speeds v – with motion restricted to a plane as we have for the rocket launch differential game above. In relative coordinates, the diagrammatic structure of the motion is as depicted in [41]. Choosing the Cartesian coordinate for motion representation, the state vector of the game with $\boldsymbol{E}$ at the origin can be characterized by its x_1, x_2 position relative to $\boldsymbol{P}$ and the angle θ between the two vehicles. Capture occurs when the distance $\|\boldsymbol{PE}\|_2$ between the pursuer and the evader becomes less than a specified radius.

The relative equations of motion, going by [41], is

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{pmatrix} = \begin{pmatrix} -v_e + v_p \cos x_3 + w_e x_2 \\ v_p \sin x_3 - w_e x_1 \\ w_p - w_e \end{pmatrix}. \tag{C.1}$$

We adopt specialization to a case where the two vehicles only possess a unit velocity and unit maximum turn rates. Here, as Merz notes, if the initial velocities are parallel such as $x_3 = 0$, then the equations of relative motion imply that $\boldsymbol{E}$ can be separated from $\boldsymbol{P}$ forever by the initial radial separation if it replicates $\boldsymbol{P}$'s strategy. A characterization of the barrier requires the separate game analysis of [29, 38]. This terminal surface possesses a closed-form solution and we refer readers to the treatment by [38]. The numerical comparison below concerns the fixed-horizon viscous Cauchy problem and an independent discretization of that same equation.

The backward reachable tube that consists of the paths taken by the trajectories of either player is defined as in the rockets pursuit-evasion game so that we have

$$\mathcal{T} = \{x \in \mathcal{X} : x_1^2 + x_2^2 \leq r^2\}, \tag{C.2}$$

where again r is the capture radius. The target is a cylinder because its definition excludes heading. A scalar initial condition represents this set through its zero sublevel set; the bounded payoff used for the recomputation is specified in Section 3(b). It is represented as shown in [41].

For a detailed treatment of the barrier surface, we refer readers to a proper analysis as elucidated in [41]. Here, we focus on the construction of the BUP. The set of states that constitute the useable part and its boundary are respectively a function of the implicit surface function representation $\ell : [0, T] \times \mathcal{X} \to \mathbb{R}$ so that for a $t \in [0, T]$, where $T > 0$ is

$$\mathcal{T} = \{x \in \mathcal{X} | \ell(0, x) \leq 0\} \tag{C.3}$$

$$R([0, t], \mathcal{T}) = \{x \in \mathcal{X} | \ell(t, x) \leq 0\}, \tag{C.4}$$

A separate single-vehicle example uses $\dot{x}_1 = \cos\theta$, $\dot{x}_2 = \sin\theta$, and $\dot{\theta} = u$, with $|u| \leq 1$ and a pose target that includes heading. Its elapsed Hamiltonian is $\boldsymbol{H} = -p_1 \cos\theta - p_2 \sin\theta + |p_3|$. Figure 2 compares the full residual computation with the independent same-viscosity reference. For a

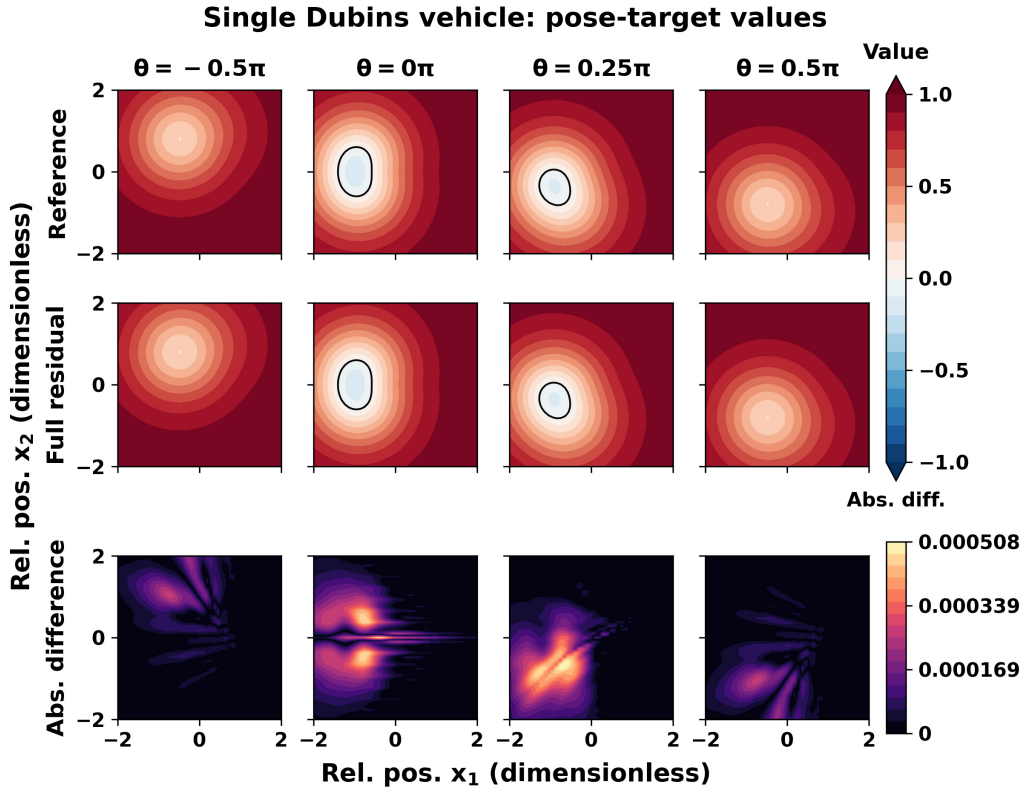


Figure 2: Single-Dubins pose-target values at four headings, $T = 1$, and $\delta = 0.08$. Rows show the independent finite-difference reference, the full residual computation, and their absolute difference. Coordinates and payoff are dimensionless. Black contours mark zero where present.

viscous reachable tube in elapsed time, the stopping formulation uses

$$\ell_t + \max\{0, \boldsymbol{H}(x, D\ell)\} = \frac{\delta}{2}\Delta\ell. \tag{C.5}$$

Optimizing the physical-time dynamics gives the Hamiltonian $\boldsymbol{H}_{\text{phys}} = \max_{w_e} \min_{w_p} p \cdot f$:

$$\boldsymbol{H}_{\text{phys}}(\boldsymbol{x}, p) = p_1(v_p \cos x_3 - v_e) + p_2(v_p \sin x_3) + w|p_1 x_2 - p_2 x_1 - p_3| - w|p_3|. \tag{C.6}$$

Since we are concerned with the special case that the linear and angular speeds are equal, we set $v_e = v_p = w \triangleq +1$ in the foregoing so that the Hamiltonian, in the final analysis is

$$\boldsymbol{H}_{\text{phys}}(\boldsymbol{x}, p) = p_1(\cos x_3 - 1) + p_2(\sin x_3) + |p_1 x_2 - p_2 x_1 - p_3| - |p_3|. \tag{C.7}$$

For elapsed backward time, the numerical Cauchy problem uses $\boldsymbol{H} = -\boldsymbol{H}_{\text{phys}}$. For unit bounds this is

$$\boldsymbol{H}(\boldsymbol{x}, p) = p_1(1 - \cos x_3) - p_2 \sin x_3 - |p_1 x_2 - p_2 x_1 - p_3| + |p_3|. \tag{C.8}$$

The recomputed terminal-horizon fields use this Hamiltonian without a stopping operator. A reachable tube requires the separate formulation in Section 2.

The independent-reference comparison is shown in Figure 3; Table 1 reports its refinement checks.

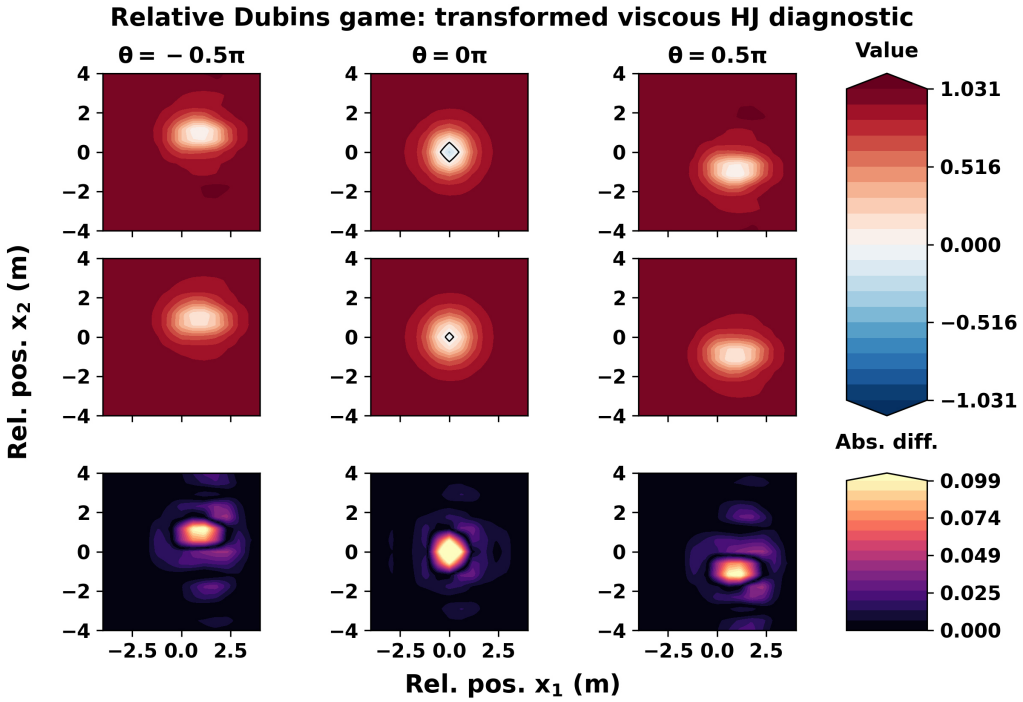


Figure 3: Relative-Dubins terminal-horizon values at $T = 1$ and $\delta = 0.08$. Rows show the independent finite-difference reference, direct integration of the full transformed equation, and their absolute difference. Columns select three headings. Black contours mark zero; coordinates and payoff are dimensionless. The comparison uses the same Hamiltonian, initial condition, and viscosity in both solvers.

(b) The double integrator

Here, we analyze a time-optimal control problem to determine what admissible control[1] can "transport" the system under consideration to a desired "origin" in the shortest possible time. We consider the double integral plant [6, 59] as an illustrative example of our objective, which is to compute the points in the state space that can reach the origin in *finite-time* under the influence of a time-optimal controller.

We shall leverage standard necessary conditions from the principle of optimality [5] to obtain a time-optimal feedback control design; introduce the notion of isochrones and switching surfaces; and discuss the analytic and approximate solutions (with our library) to the time-optimal control problem for a double integrator. This time-optimal discussion supplies geometric context. The recomputed numerical experiment uses a fixed horizon and a bounded target-ball payoff, as specified in Section 3(b).

(i) Dynamics and Problem Setup

The double integrator is controllable, so that open-loop strategies may be employed in driving specific states to the origin in finite time [59]. The plant has the following second-order dynamics

$$\ddot{\boldsymbol{x}}(t) = \boldsymbol{u}(t) \tag{C.9}$$

[1] A control law is admissible when its range belongs in the admissible input set where it is bounded.

and admits bounded control signals $|\boldsymbol{u}(t)| \leq 1$ for all time t. After a change of variables,we have the following system of first-order differential equations

$$\dot{\boldsymbol{x}}_1(t) = \boldsymbol{x}_2(t), \qquad \dot{\boldsymbol{x}}_2(t) = \boldsymbol{u}(t), \qquad |\boldsymbol{u}(t)| \leq 1. \tag{C.10}$$

The *reachability problem that we consider is to address the question of what states can reach a certain point (here, the origin) in a transient manner*. That is, we would like to find point sets on the state space, at a particular time step, such that we can bring the system to the equilibrium, $(0, 0)$.

(ii) Time-optimal control scheme

This is an $\boldsymbol{H}$-minimal control problem whereupon we must find the control law that minimizes the Hamiltonian

$$\boldsymbol{H}_{\text{phys}}(\boldsymbol{x}, p, \boldsymbol{u}) = p_1 \boldsymbol{x}_2 + p_2 \boldsymbol{u}. \tag{C.11}$$

Minimization gives $\boldsymbol{H}_{\text{phys}} = p_1 \boldsymbol{x}_2 - |p_2|$. The elapsed-time Hamiltonian and the minimizing physical control, for $p_2(t) \neq 0$, are

$$\boldsymbol{H}(\boldsymbol{x}, p) = -p_1 \boldsymbol{x}_2 + |p_2|, \qquad \boldsymbol{u}^*(t) = -\operatorname{sign}(p_2(t)). \tag{C.12}$$

Any admissible control minimizes the instantaneous Hamiltonian when $p_2(t) = 0$.

For the co-states in question, suppose that their initial values (for constants k_1 and k_2) are $p_1(t_0) = k_1$ and $p_2(t_0) = k_2$, only four candidates can serve as time-optimal control sequences i.e., $\{[+1], [-1], [+1, -1], [-1, +1]\}$. On each constant-control segment, $t \in [t_0, t_f]$, write $\boldsymbol{u}(t) = k \in \{-1, +1\}$ so that for initial conditions $\boldsymbol{x}_1(t_0) = \boldsymbol{\xi}_1$ and $\boldsymbol{x}_2(t_0) = \boldsymbol{\xi}_2$, it can be verified that the state trajectories obey the relation

$$\boldsymbol{x}_1(t) = \boldsymbol{\xi}_1 + \frac{1}{2} k \left(\boldsymbol{x}_2^2 - \boldsymbol{\xi}_2^2 \right), \text{ for } t - t_0 = k \left(\boldsymbol{x}_2(t) - \boldsymbol{\xi}_2 \right). \tag{C.13}$$

Figure 4 shows the full residual double-integrator fields at four elapsed horizons. The numerical protocol and sensitivity checks are given in Section 3(b). Equation (C.13) describes each constant-control segment; concatenating segments with opposite controls gives the switching trajectories of the classical time-optimal problem [6, 59].

The numerical refinement and independent-reference comparisons are reported in Table 1.

(c) A normalized rockets benchmark

The rocket-launch examples of Dreyfus [16] and Molu [44] motivate a normalized controlled benchmark. Its numerical definition is the following normalized relative-state system:

$$\dot{x}_1 = a \cos\theta + u_e x_1, \quad \dot{x}_2 = a \sin\theta + a + u_p x_1 - g, \quad \dot{\theta} = u_p - u_e. \tag{C.14}$$

Here $\theta \in \mathbb{S}^1$ is an independent relative-heading state, and $u_e, u_p \in [-1, 1]$ are rate controls. The evader maximizes the payoff and the pursuer minimizes it. With $p = D\boldsymbol{v}$, collection of the control terms gives

$$p \cdot f = a p_1 \cos\theta + (a + a \sin\theta - g) p_2 + (p_1 x_1 - p_3) u_e + (p_2 x_1 + p_3) u_p. \tag{C.15}$$

Thus elapsed backward time uses

$$\boldsymbol{H} = -\max_{u_e} \min_{u_p} p \cdot f = -a p_1 \cos\theta + (g - a - a \sin\theta) p_2 - |p_1 x_1 - p_3| + |p_2 x_1 + p_3|. \tag{C.16}$$

The recomputation sets $a = 1$, $g = 2$, target radius $r = 1.5$, viscosity $\delta = 0.08$, and horizon $T = 0.5$. Figure 1 compares the full residual solution with the independent reference at the same viscosity. The historical physical-unit rocket illustrations are given in [44]. The relative-Dubins comparison in Figure 3 supplies the principal pursuit–evasion example in this appendix.

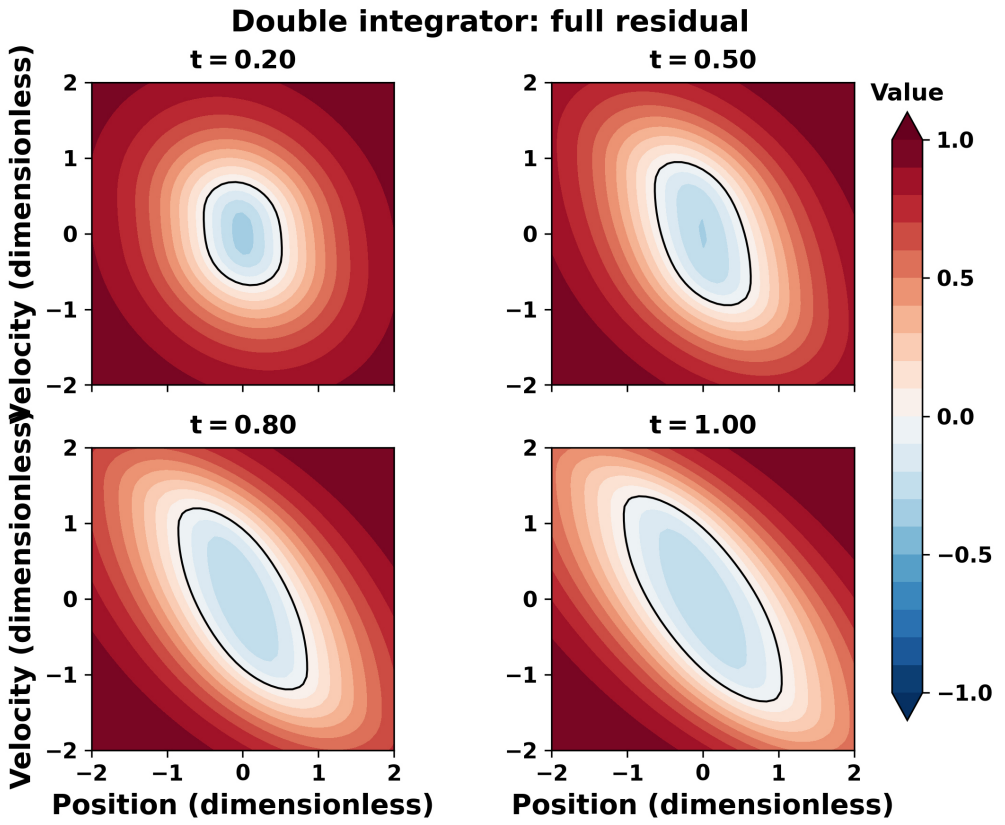


Figure 4: Full residual double-integrator values at elapsed horizons $t = 0.20, 0.50, 0.80$, and 1.00, with $\delta = 0.08$. Black contours mark the zero level set. Position, velocity, time, and payoff are dimensionless in this normalized example.

(d) Population-scale evaluation for starling-inspired dynamics

We take inspiration from natural swarms (see Fig. 5), particularly the murmuration of European starlings (*sturnus vulgaris*). In these settings, local flocks within large murmurations maintain an anisotropic formation based on a topological interaction, regardless of sparsity of birds on a phase space [8]. Thus, intra- and inter-flock collisions are avoided and attacks are fended off [4]. Approximating the viscosity solutions of nonconvex Hamilton-Jacobi partial differential equations with our quasilinearization scheme and importance sampling, the Hamiltonian, control laws, and strategies that govern the transient behaviors of many systems that possess structural subsystems with unique nearest neighbor properties may be computed.

Through empirical [4, 7, 8, 22, 58] and theoretical findings [31], evidence now abounds that in certain natural species that exhibit collective behavior, convergence and group cohesion is based on simple topological interaction rules that they employ to keep a tab on one another in *local flocks* for collision avoidance, preserving density and structure in an anisotropic formation, and exhibiting flock splitting, vacuole, cordon, and flash expansion isotropically [20]. This aids these animals in emerging an eye-pleasing local anisotropic synchrony, which taken together among possibly hundreds of thousands of local interactions[2] [20], keep these animals whirling, swooping, and flying in isotropic formations [4]. Thus, individual agents aggregate into substructures within the

[2]It has been reported that no birds fly together with greater coordination and complexity than European starlings, with murmurations counting upwards of 750,000 individual birds!

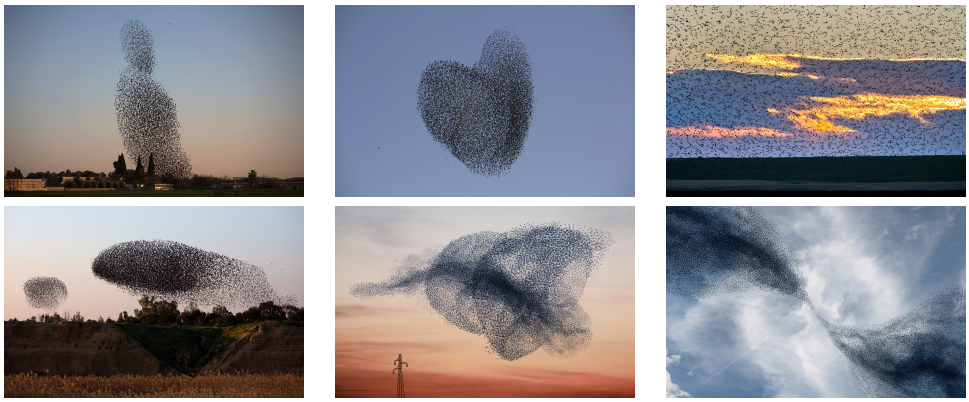


Figure 5: Photographs illustrate collective behaviors of European starlings (*Sturnus vulgaris*); they are not outputs of the numerical solver. From the top-left and clockwise. (i) A starlings flock rises into the air, in a dense structure (Reuters/Amir Cohen). (ii) Starlings migrating over an Israeli village (AP Photo/Oded Balilty). (iii) Starlings feeding on laid seeds in the ground in Romania. (iv) Two flocks of migrating starlings (Menahem Kahana/AFP/Getty Images). (v) A concentric conical formation of starlings (Courtesy of The Gathering Site.). (vi) Splitting and joining of a flock of starlings.

overall system, and overall group motion is synergized via local topological interactions so that a stable global heading and cohesion [31] is preserved.

(i) Notations and Background

A few mathematical notations, conventions, and taxonomy used throughout this section are in order at this juncture. Capital and lower-case Roman letters are matrices and vectors respectively. Exceptions: time variables such as t, t_0, t_f, T are real numbers throughout. Calligraphic letters are sets. Exception: the HJ equation's solution (shortly introduced) is the set $\Omega \in \mathbb{R}^n$ in which agents move. We work in a multi-agent system context where individual agents self-organize into phases or regions $\mathcal{S}$ which are in turn members of a union of multiple regions $\mathcal{C}$. Note that every $\mathcal{S} \subseteq \mathcal{C}$ and all members of $\mathcal{C}$ are disjoint from one another i.e., $\mathcal{S}_i \cup \mathcal{S}_j = \emptyset$ for any $i \neq j$. The total number of elements in $\mathcal{S}$ is denoted $[\mathcal{S}]$, and we denote by int Ω the interior of Ω. The closure of Ω is $\bar{\Omega}$. We let $\delta\Omega\,(:= \bar{\Omega} \backslash \text{int}\ \Omega)$ be the boundary of Ω.

Structural homogeneity of starlings' motion in every region $\mathcal{S}_i \in \mathcal{C}$ for $i = 1, \cdots, [\mathcal{C}]$ applies; this is enforced by introducing an external disturbance on the zeroth-index agent (this aids compactness of the zero levelset of an $\mathcal{S}_i$ as we will introduce shortly). Hence, each safety verification episode can be characterized as a pursuit *game*, $\mathit{\Gamma}$. And by a game, we do not necessarily refer to a single game, but rather a *collection of games*, $\mathit{\Upsilon} = \{\mathit{\Gamma}_1, \cdots, \mathit{\Gamma}_g\}$. Such a game terminates when *capture* occurs, that is the distance between players falls below a predetermined threshold. As in previous examples, each player in a game shall constitute either a pursuer ($\boldsymbol{P}$) or an evader ($\boldsymbol{E}$). Let the cursory reader not interpret $\boldsymbol{P}$ or $\boldsymbol{E}$ as controlling a single agent. In our setup, we are poised with several pursuers (e.g. Falcons) or evaders (starlings). However, when $\boldsymbol{P}$ or $\boldsymbol{E}$ governs the behavior of but one bird, these symbols will denote the bird itself. An evading bird in a region $\mathcal{S}_i$ has a state notation $\boldsymbol{x}_a^i$ (read: the state of bird a in region i). A state $\boldsymbol{x}_a^i$ has linear velocity components, $\boldsymbol{x}_{a_1}^i, \boldsymbol{x}_{a_2}^i$, and heading $\boldsymbol{x}_{a_3}^i := w_a^i$. When we must distinguish an bird $\boldsymbol{x}_a^i \in \mathcal{C}_x$ from some other bird e.g. in another multiphase $\mathcal{C}_y$, we shall write ${}^x\boldsymbol{x}_a^i$ and ${}^y\boldsymbol{x}_a^i$ respectively. Given the various possibilities of outcomes, the "best" outcome is resolved by a *payoff*, $\mathit{\Phi}$, whose extremal over a time interval will constitute a *value*, $\boldsymbol{v}$[3]. We adopt the terminology of Isaacs [29] so that if the payoff for a game is finite we shall have a *game of kind* (a qualitative game); and for a game with a continuum

[3]The functional $\boldsymbol{\Phi}$ may be considered a functional mapping from an infinite-dimensional space to the space of real numbers.

of payoffs we shall have *a game of degree* (quantitative games). The *strategy* executed by $\boldsymbol{P}$ or $\boldsymbol{E}$ during a game shall be denoted by $\alpha \in \mathcal{A}$ (resp. $\beta \in \mathcal{B}$).

The many interacting subsystems under consideration employ (i) natural units of measurements that are the same for all birds; (ii) kinematics with same linear speeds but with a capacity for orientation changes; (iii) inter-region interaction occurs within unique and distinct state space manifolds; and by birds maneuvering their direction, a kinematic alignment is obtained with other regions; (iv) region-to-region interaction occurs when a pursuer is within a threshold of capturing any bird in a region; (v) the interaction among respective regions is described by the time-evolution of an interface, which is the zero-level set of the objective functional of the respective local subgroups.

(ii) Murmuration as a Levelsets Fronts Equation

The key idea is that when the multi-bird problem has separable structure, one can resolve the associated HJ equations in a numerically consistent manner by considering the interface of these separable structures, i, as evolving dynamic interfaces, Γ_i, via the modified levelset equation (HJ-IVP)

$$\phi_t + \Gamma_i \boldsymbol{H}(\boldsymbol{x}, t, \phi, D\phi) = 0 \tag{C.17}$$

A reconstructed unsigned distance can describe interface geometry, but it does not distinguish the safe and unsafe sides. A control interpretation therefore requires the retained sign or value field together with a validated game formulation and numerical error bound. Gaussian source representations can approximate a declared interface equation; they do not supply those missing modeling and sign guarantees by themselves.

Popular numerical tools for simulating the evolution of interfaces in applied physics, mathematics, and computational sciences include fast marching front tracking methods [52, 53], Voronoi implicit interface methods (or VIIMs) [51, 60], discretization schemes for Hamilton-Jacobi equations [56], or multilayer volume-of-fluid methods [32] for foaming across scales. While each of these schemes has its own advantages, we will resort to levelset methods [54] in developing algorithmic efficiency for the trajectory optimization among a *collection* ($\mathcal{C}$) of *disjoint* multiple starlings *sets*, $\mathcal{S} \subseteq \mathcal{C}$. The entire collection of starlings moves over an open set $\Omega \subset \mathbb{R}^n$. When subgroups $\{\mathcal{S}_i\}_{i=1}^{i=k} \subset \mathcal{C}, k < n$ of aerial starlings must traverse a narrow opening, or temporarily break apart to avoid collisions, for example, we expect topological changes to occur without explicit surgery. Furthermore, group geometric formation and cohesion are achieved by local changes to subgroups' collective heading and speed. This is a challenging problem numerically and we look to natural behaviors in animals and multiphase simulations for inspiration. In our problem construction, a *finite set* of individual starlings self-organize into local structural groups (which we refer to as subgroups, $\mathcal{S}$); subgroups interact based on nearest neighbor rules so that effective group consensus is dictated by interactions among separate subgroups [31].

Throughout, our theater of operations involve multiple aerial Dubins vehicles [17] with an extra dimension along the vertical position to describe the 4-dimensional motion of birds. We treat bird motions as kinematic models that possess linear and angular speeds as state variables. The safety verification optimization problem is played as a game of multiple vehicles in a four-dimensional space. As we are dealing with multiple starlings, objectives such as collision avoidance, starlings' spatial separation, overall group coherence become paramount. At issue is continually computing the points set that belong on the *reachable set* boundary as the game advances forward in time. These reachable sets are those state space subsets where starlings may collide into one another, hit obstacles, lose topological integrity or miss group coherence behavior.

(iii) Related Works

As all starlings move in $\Omega \subset \mathbb{R}^n$, we want a stable and safe numerical algorithm that correctly represents dynamic interface boundary conditions, accurately represents kinematics, whilst sensitive enough to rapidly interpret subgroup topological and structural changes. Tsitsiklis [56]

worked under the restrictive assumption that HJ's running cost is independent of the control laws' update – providing an $O(n/p)$ parallel algorithm that resolved the discretized HJ-Hamiltonian for p processors on n grid points provided that $p = O(\sqrt{n}/\log n)$. In front tracking methods, a Lagrangian geometric representation techniques track the surfaces between separate structures with mechanisms such as triple point junctions which are endowed with shared nodes so that members of the front set are updated as time evolves [33]. In applied mechanics and microfluidics, methods such as volume-of-fluid [3] resolve the interaction among multiple phases that are separated by thin boundaries with volume fraction fields for each region or unique functions in levelset methods – leading to coalescence prevention among multiple regions. However, they come at a computational cost of $\mathcal{O}(N_{regions}N_{cells})$. To improve its scalability, Karnakov and colleagues [32] compactly stored many fields thereby keeping needed scalar fields constant and independent of regions to simulate.

(iv) Problem Formulation

We resolve local payoff extremals, $\{\Phi_1, \cdots, \Phi_{n_f}\}$. as a state space partition induced by an aggregation of desired collective behavior from local flocks' values $\{\boldsymbol{v}_1, \cdots, \boldsymbol{v}_{n_f}\}$. Suppose that the local control laws are properly coordinated, the region of the state space across which their coordinated influence might be exerted constitute a larger e.g. *manipulability volume* for a dexterous kinematic task. We now formalize definitions that will aid the modularization of the problem into manageable forms.

Definition C.1 (Neighbors of an Agent)**.** We define the neighbors $\mathcal{N}_i(t)$ of agent i at time t as the set of all agents that lie within a predefined radius, r_i.

Definition C.2. We define a *flock*, F, consisting of agents labeled $\{1, 2, \cdots, n_a\}$ as a collection of agents within a phase space $(\mathcal{X}, T)$ such that all agents within the flock interact with their nearest neighbors in a topological sense.

Remark C.1. Every agent within a flock has similar dynamics to that of its neighbor(s). Furthermore, agents travel at the same linear speed, v; climb/dive rate (the control) $u_z \in [-\gamma_{\max}, \gamma_{\max}]$, where $\gamma_{\max} \in \mathbb{R}$; the angular headings, w, however, may be different between agents, seeing we are dealing with a many-bodied system. Each agent's continuous-time dynamics, $\dot{\boldsymbol{x}}^{(i)}(t)$, evolves as

$$\begin{bmatrix} \dot{\boldsymbol{x}}_1^{(i)}(t) \\ \dot{\boldsymbol{x}}_2^{(i)}(t) \\ \dot{\boldsymbol{x}}_3^{(i)}(t) \\ \dot{\boldsymbol{x}}_4^{(i)}(t) \end{bmatrix} = \begin{bmatrix} v(t)\cos \boldsymbol{x}_4^{(i)}(t) \\ v(t)\sin \boldsymbol{x}_4^{(i)}(t) \\ u_z(t) \\ \langle w^{(i)}(t)\rangle_r \end{bmatrix}, \quad \langle w^{(i)}(t)\rangle_r \quad = \frac{1}{1+n_i(t)}\left(w^{(i)}(t) + \sum_{j\in\mathcal{N}_i(t)} w_j(t)\right) \tag{C.18}$$

for agents $i = \{1, 2, 3, ..., n_a\}$, where t is the continuous-time index, $n_i(t)$ is the number of agent i's neighbors at time t, $\mathcal{N}_i(t)$ denotes the sets of labels of agent i's neighbors at time t, and $\langle w^{(i)}(t)\rangle_r$ is the average orientation of agent i and its neighbors at time t. Note that for a game where all agents share the same constant linear speed and heading, and $u_z = 0$, (C.18) reduces to the dynamics of a Dubins' vehicle in absolute coordinates with $-\pi \leq w^{(i)}(t) < \pi$. The averaging over the degrees of freedom of other agents in (C.18) is consistent with *mean field theory*, where the effect of all other agents on any one agent is an approximation of a single averaged influence.

Definition C.3 (Payoff of a Flock)**.** To every flock F_j (with a finite number of agents n_a) within a murmuration, $j = \{1, 2, \cdots, n_f\}$, we associate a payoff, $\boldsymbol{\Phi}_j$, that is the union of all respective agent's payoffs for expressing the outcome of a desired kinematic behavior.

Flock F_j and F_k within a murmuration, $F_j \cup F_k \cup F_l \cdots$ are separated by *partitions*, or *interfaces*, $\Gamma_{jk}, \Gamma_{kl}, \cdots$. This interface may be implicitly represented as a signed distance function $\Phi(\boldsymbol{x})$ which

is negative on the interior of each flock, and zero on the edges. The zero-level set (i.e., $\Phi(\boldsymbol{x}) = 0$) corresponds to the interface $\boldsymbol{v}$ [52]. As the system evolves over time, F_j's interface (zero-level set) motion can be parameterized by time, so that the flow field $\boldsymbol{v}(\boldsymbol{x}, t)$ is equivalent to the solution of the Cauchy-type Hamilton Jacobi partial differential equation [10, 18]:

$$\boldsymbol{v}_t + v_j|\nabla \boldsymbol{v}_j| = 0, \quad j = 1, \cdots, n_f, \tag{C.19}$$

where v_j is the flow speed for F_j. Equation (HJ-IVP) is the level set equation [54].

In the sentiment of [42], we say the zero sublevel set of $g(\cdot)$ i.e., $\mathcal{L}_0$ in (2.15*a*) is the *target set* in the phase space $\Omega \times \mathbb{R}$ for a backward reachability problem [39]. This target set[4] can represent the failure set, regions of danger, or obstacles to be avoided e.t.c. in the vectogram. And the *robustly controlled backward reachable tube* for $\tau \in [-T, 0]$[5] is the closure of the open set $\mathcal{L}([\tau, 0], \mathcal{L}_0)$ in (2.15*b*).

Each agent within a flock interacts with a fixed number of neighbors, n_c, within a fixed topological range, r_c. This topological range is consistent with findings in collective swarm behaviors and it reinforces *group cohesion* [4]. However, we are interested in *robust group cohesion* in reachability analysis. Therefore, we let a pursuer, $\boldsymbol{P}$, with a worst-possible disturbance attack the flock, and we take it that flocks of agents constitute an evading player, $\boldsymbol{E}$. Returning to (C.18), for a single flock, we now provide a sketch for the HJI formulation for a heading consensus problem.

(v) Framework for Separated Payoffs.

Suppose that a murmuration's global heading is predetermined and each agent i within each flock, F_j, $(j = \{1, \cdots, n_f\})$ in the murmuration has a constant linear velocity, v^i. An agent's orientation is its control input, given by the average of its own orientation and that of its neighbors. Instead of metric distance interaction rules that make agents very vulnerable to predators [4], we resort to a topological interaction rule. With metric distance rules, we will have to formulate the breaking apart of value functions that encode a consensus heading problem in order to resolve the extrema of multiple payoffs; which is typically what we want to prevent in real-world autonomous tasks.

What constitutes an agent's neighbors are computed based on empirical findings and studies from the lateral vision of birds and fishes [4, 22, 31] that provide insights into their anisotropic kinematic density and structure. Importantly, starlings' lateral visual axes and their lack of a rear sector reinforces their lack of nearest neighbors in the front-rear direction. As such, this enables them to maintain a tight density and robust heading during formation and flight.

Each agent within a flock F_j interacts with a fixed number of neighbors, n_c, within a fixed topological range, r_c. The topological range can be set as the distance between the labels of agents in a flock. This topological range is consistent with findings in collective swarm behaviors and it reinforces *group cohesion* [4]. However, we are interested in *robust group cohesion* in reachability analysis. Therefore, we let a pursuer, $\boldsymbol{P}$, with a worst-possible disturbance attack the flock, and we take it that flocks of agents constitute an evading player, $\boldsymbol{E}$.

(vi) Global Isotropy via Local Anisotropy.

Structural anisotropy is not merely an effect of a preferential velocity in animal flocking kinematics but rather an explicit effect of the anisotropic interaction character itself: agents choose a mutual position on the state space in order to maximize the sensitivity to changes in heading and speed of neighbors as the neighbors' anisotropy is optimized via vision-based collision avoidance characteristically unrelated to the eye's structure [4].

An individual relative-state model supplies a local description of an agent under attack. Coordinating such descriptions into a jointly controlled flock requires an additional coupling model. For equal speeds and matched headings, identical admissible steering signals can preserve the initial relative separation in the corresponding kinematic model. This observation alone does not establish that a differential-game barrier is closed. A barrier characterization requires a separate game analysis.

[4]Note that the target set, $\mathcal{L}_0$, is a closed subset of $\mathbb{R}^n$ and is in the closure of Ω.
[5]The (backward) horizon, $-T$ is negative for $T > 0$.

A Voronoi partition can organize local flock descriptions, but it does not prove that their assembled value fields solve a joint reachability problem. The following dynamics specify an individual relative-coordinate model; global composition and its numerical consistency remain additional requirements.

For the numerical individual model, write $\boldsymbol{x} = (\boldsymbol{x}_1, \boldsymbol{x}_2, z, \theta)$. The physical-time dynamics are

$$\dot{\boldsymbol{x}} = \begin{bmatrix} -1 + \cos\theta + w_e \boldsymbol{x}_2 \\ \sin\theta - w_e \boldsymbol{x}_1 \\ u_p^z - u_e^z \\ w_p - w_e \end{bmatrix}, \tag{C.20}$$

with $w_e, w_p \in [-1, 1]$ and $u_e^z, u_p^z \in [-\gamma, \gamma]$, where $\gamma = 0.5$. The altitude state satisfies $z \in \mathbb{R}$, and $\theta \in \mathbb{S}^1$. The numerical model uses an independent admissible rate w_e as a surrogate for neighbor-dependent averaged steering. The population-query experiment evaluates this individual controlled field; it does not simulate the interacting neighbor law.

For an evader maximizing and a pursuer minimizing the payoff, the vertical contribution to the physical Hamiltonian is $\gamma_e|p_z| - \gamma_p|p_z|$. Equal vertical bounds cancel it. The altitude-independent initial condition then gives an altitude-independent solution, whose three-dimensional elapsed Hamiltonian is (C.8). The reversal $\boldsymbol{H} = -\boldsymbol{H}_{\text{phys}}$ is applied once when forming the elapsed-time Cauchy equation. Sampling altitude in $[-1, 1]$ specifies the query distribution and imposes no altitude state constraint.

For context, in absolute coordinates, the equation of motion for *free agents* is

$$\begin{bmatrix} \dot{\boldsymbol{x}}_1^{(i)_j}(t) \\ \dot{\boldsymbol{x}}_2^{(i)_j}(t) \\ \dot{\boldsymbol{x}}_3^{(i)_j}(t) \\ \dot{\boldsymbol{x}}_4^{(i)_j}(t) \end{bmatrix} = \begin{bmatrix} v_e^{(i)_j}(t) \cos \boldsymbol{x}_4^{(i)_j}(t) \\ v_e^{(i)_j}(t) \sin \boldsymbol{x}_4^{(i)_j}(t) \\ u_z^{(i)_j}(t) \\ \langle w_e^{(i)_j}(t) \rangle_r \end{bmatrix}. \tag{C.21}$$

(vii) Flock Motion from Aggregated Value Functions.

We introduce the union operator i.e., $\cup$ below as an aggregation symbol since the respective payoffs of each agent in a flock may be implicitly or explicitly constructed[6] – when it is implicitly represented, say from a signed distance function, we shall aggregate the payoff of agents 1 and 2 as

$$\cup\{\Phi_1(\boldsymbol{x}, t), \Phi_2(\boldsymbol{x}, t)\} \equiv \Phi_1(\boldsymbol{x}, t) \cup \Phi_2(\boldsymbol{x}, t) = \min(\Phi_1(\boldsymbol{x}, t), \Phi_2(\boldsymbol{x}, t)) \tag{C.22}$$

otherwise, other appropriate arithmetic or logical operation shall apply.

Remark C.2 (Proposed aggregation of flock values)**.** A pointwise minimum combines scalar functions whose zero sublevel sets represent a union. For separately computed fields, the proposed display aggregate is

$$\Phi_{\text{union}}(\boldsymbol{x}, t) = \min_j \Phi_j(\boldsymbol{x}, t), \qquad \{\Phi_{\text{union}} \le 0\} = \bigcup_j \{\Phi_j \le 0\}. \tag{C.23}$$

The set identity follows from the definition of a minimum. It does not imply that a union or minimum of Hamiltonians governs a coupled flock game. Optimizing separately and then taking a minimum need not commute with optimizing a shared strategy. A joint model requires a specified payoff, state, control coupling, and dynamic-programming argument.

For each declared individual model, its physical-time game Hamiltonian has the control form

$$\boldsymbol{H}_{\text{phys}}^{(i)}(t, \boldsymbol{x}, \boldsymbol{p}) = \max_{\boldsymbol{u} \in \mathcal{U}_i} \min_{\boldsymbol{w} \in \mathcal{W}_i} \boldsymbol{p} \cdot f^{(i)}(t, \boldsymbol{x}, \boldsymbol{u}, \boldsymbol{w}). \tag{C.24}$$

The preceding dynamics specify candidate individual vector fields. Their coordinate conventions and control bounds must agree with the chosen numerical model before this formula defines a

[6] In resolving the zero-level sets of HJ value functions, it is typical to represent the payoff's surface as the isocontour of some function (usually a signed distance function).

computed value. An individual Hamiltonian is not automatically a mechanical energy, and a set union of its scalar values is not a scalar differential operator.

Remark C.3 (Scope of the population model)**.** The illustrations combine low-dimensional agent or flock calculations. They do not establish a Hamilton–Jacobi equation on the full joint population state. Identifying a population game with those aggregates would require a decomposition theorem or a separately verified approximation bound. Neither follows from signed-distance initialization or equal agent speeds.

(viii) Vacuole nucleation diagnostics

A hole in a displayed zero sublevel set can serve as a geometric marker of a vacuole. Predator entry and a local Hamiltonian threshold alone do not force that topology. Let $\mathcal{L}_h(t)$ denote a finite planar cubical approximation of the displayed sublevel set.

Definition C.4 (Euler characteristic of a displayed sublevel set)**.** For consistent vertex, edge, and face counts V, E, F of $\mathcal{L}_h(t)$, define

$$\chi_h(t) = V - E + F = \beta_0(t) - \beta_1(t). \tag{C.25}$$

The counts β_0 and β_1 denote connected components and independent holes of that planar complex. Grid connectivity and boundary treatment form part of the definition [21].

Remark C.4 (Local thresholds and topology)**.** A condition such as $\boldsymbol{H}_f(\boldsymbol{x}^*, \boldsymbol{p}^*) = 0$ does not identify a critical point of $\boldsymbol{v}$, a Morse saddle, or a change in a global sublevel set. A smooth Morse-theoretic argument would need these additional hypotheses and a specified parameter crossing. Even an index-one handle can merge two components rather than create a hole. A vacuole interpretation therefore requires spatial evidence of an enclosed complementary region, rather than a local Hamiltonian test alone.

(ix) Flock splitting diagnostics

Remark C.5 (Local motion and global disconnection)**.** A negative attacked-agent Hamiltonian can describe local motion under the model's sign convention. It is neither a necessary nor a sufficient condition for a displayed reachable set to become disconnected. Global disconnection depends on the entire field and domain. A rise in the measured component count is a proposed fragmentation marker; linking it to a particular control threshold or game equilibrium requires a separate argument.

(x) Cordon geometry

Definition C.5 (Annular display region)**.** For $0 < r_{\text{in}} < r_{\text{out}}$, define the horizontal annular region

$$\mathcal{C} = \{\boldsymbol{x} : r_{\text{in}} \le \|\boldsymbol{x}_{1:2}\| \le r_{\text{out}}\}. \tag{C.26}$$

Remark C.6 (Topology does not certify a barrier)**.** An annular displayed component has one hole, but that geometry alone gives no invariant barrier or finite-time capture guarantee. Such conclusions require the appropriate dynamical boundary condition and value-error control. In particular, the subset $\{\boldsymbol{x} \in \mathcal{C} : \|\boldsymbol{x}_{1:2}\| < r_{\text{in}}\}$ is empty, so a sign condition on that subset supplies no information about the interior. A cordon interpretation must also verify which agents or regions the displayed loop encloses.

(xi) Flash expansion diagnostics

Remark C.7 (Trajectory speed and sublevel-set motion)**.** A horizontal speed bound v_e bounds an individual free trajectory's horizontal displacement by $v_e \Delta t$ over an interval of length Δt. It does not by itself bound the motion of a relative-game sublevel set, which also depends on pursuer

motion, controls, the target, and time orientation. A reachable-tube rate requires an additional argument. Changes in the displayed radius remain descriptive diagnostics.

(xii) Markers in numerical slices

Proposed geometric diagnostics use a two-dimensional slice of the four-dimensional value approximation. Fix the remaining state coordinates, a grid, a sign threshold, and a connectivity convention. Compute the resulting cubical complex, its connected components, and its enclosed complementary components. Record χ_h, β_1, and n_{comp} for each sampled time.

Remark C.8 (Observational event labels)**.** An increase in β_1 suggests a new displayed hole, and an increase in n_{comp} suggests fragmentation of the displayed set. A change in χ_h alone cannot distinguish those mechanisms. Calling a hole a cordon also requires an enclosure test. These labels describe the chosen slice and threshold; they neither classify every flock behavior nor establish a disjoint partition of the physical state space.

Grid resolution, slicing, smoothing, boundary conditions, and numerical value error can change these counts. Robust event identification requires sensitivity analysis and evidence that the relevant signs and connections persist under the numerical error budget. No such topological certification follows from the retained-source Cauchy theorem alone.

(xiii) Numerical geometry and its interpretation

The numerical slices illustrate the geometric diagnostics under a four-dimensional flock model. An annular zero sublevel set, its closure, and changes in connectivity provide compact descriptions of the displayed geometry. These observations depend on slicing, sign threshold, smoothing, and resolution; their interpretation as a global safety event requires a separate error and model-composition argument.

(e) Validation of the retained transformation source

A manufactured advection–diffusion solution isolates the effect of retaining the source. On the periodic interval $[0, 2\pi)$, choose $\delta = 0.4$, $\boldsymbol{H}(\boldsymbol{p}) = 0.7\boldsymbol{p}$, initial condition $\boldsymbol{g}(\boldsymbol{x}) = \sin \boldsymbol{x}$, and prescribed coefficient $\boldsymbol{c}(t, \boldsymbol{x}) = 1 + 0.2 \cos \boldsymbol{x} + 0.1 \sin t$. The reference solution is

$$\boldsymbol{v}^{\delta}(t, \boldsymbol{x}) = e^{-\delta t/2} \sin(\boldsymbol{x} - 0.7t). \tag{C.27}$$

The horizon is $t = 0.5$. Fourier collocation uses 128 equally spaced states, and exponential-Euler time stepping uses 4 resolutions from 100 to 800 steps. The comparison changes only the source: retain the full source or use homogeneous heat flow. Both cases use the same coefficient, initial condition, state grid, and time resolutions.

Figure 8 reports the maximum value and gradient errors at the evaluation states. With the full source, observed value-error orders range from 1.000 to 1.001. The finest value error is 1.003×10^{-4}, compared with 3.839×10^{-1} for the homogeneous heat solve. These measurements support source retention and temporal consistency on this smooth periodic problem. They do not establish a continuum error certificate, whole-space accuracy, a game benchmark, or high-dimensional computational scaling.

(i) Multidimensional manufactured verification

The multidimensional implementation is checked against

$$\boldsymbol{v}_{\mathrm{ref}}(t, \boldsymbol{x}) = e^{-\delta d t/2} \prod_{j=1}^{d} \sin(\boldsymbol{x}_j - a_j t), \qquad a_j = 0.15j, \tag{C.28}$$

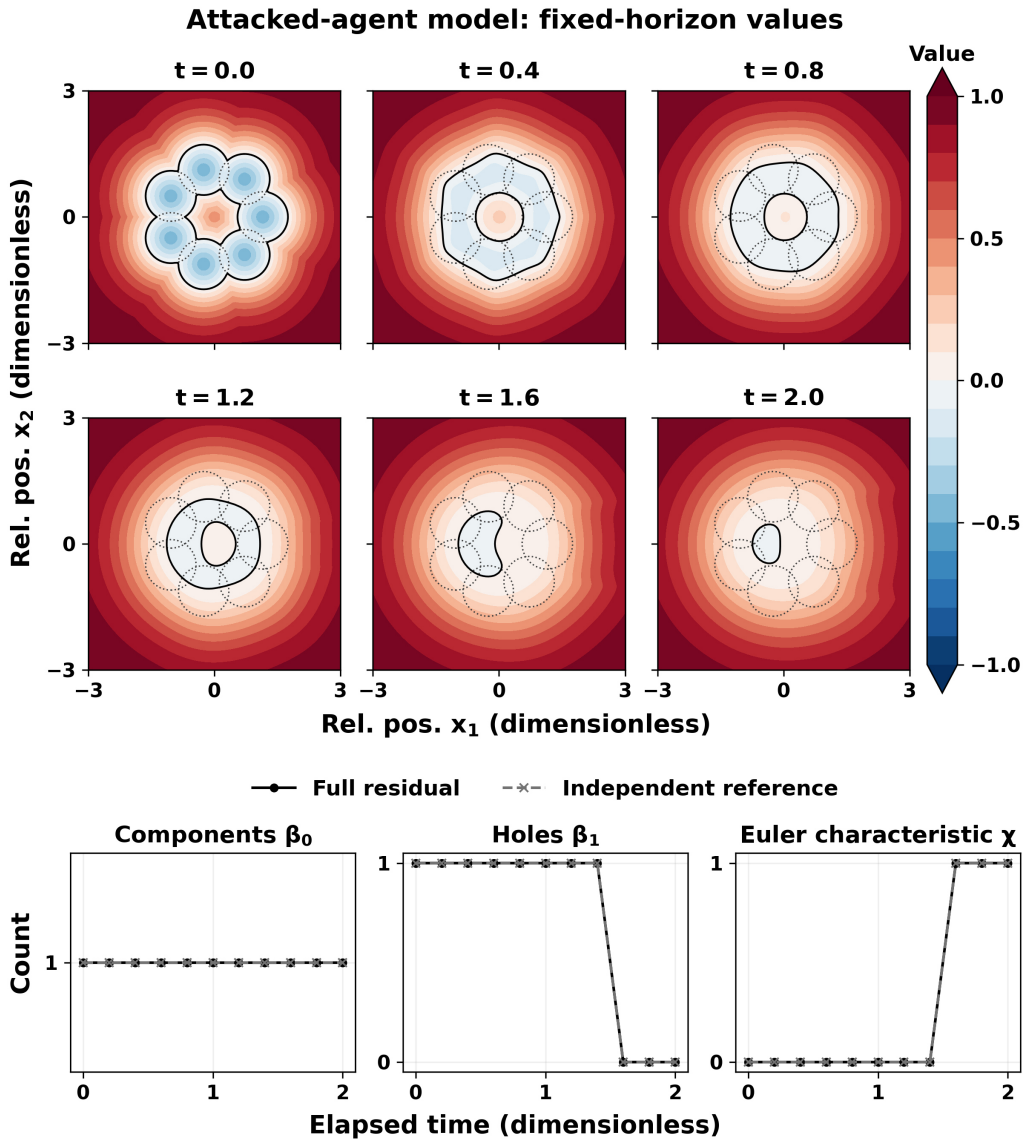


Figure 6: Numerical geometry of the flock value approximation. The upper panels show zero sublevel sets at six sampled horizons, with pursuers, capture radii, and zero contours. The lower panel displays the component count, hole count, and Euler characteristic used to describe the slice topology. These are geometric diagnostics rather than a population-wide sign certificate.

on $[-\pi,\pi)^d$, for $d=2,3$, $\delta=0.3$, and horizon 0.4. The Hamiltonian is

$$\boldsymbol{H}(t,\boldsymbol{x},\boldsymbol{p})=\sum_{j=1}^{d}a_jp_j+0.1\big(|\boldsymbol{p}|^2-|D\boldsymbol{v}_{\mathrm{ref}}(t,\boldsymbol{x})|^2\big). \tag{C.29}$$

The prescribed coefficient is (3.2), with $L_j=2\pi$. Its time and spatial derivatives are nonzero. Grid counts increase through 16, 24, 32, and 48 per coordinate, while the step counts increase through 40, 80, 160, and 320. Separate fixed-grid time-halving tests verify second-order temporal refinement of the transformed solver.

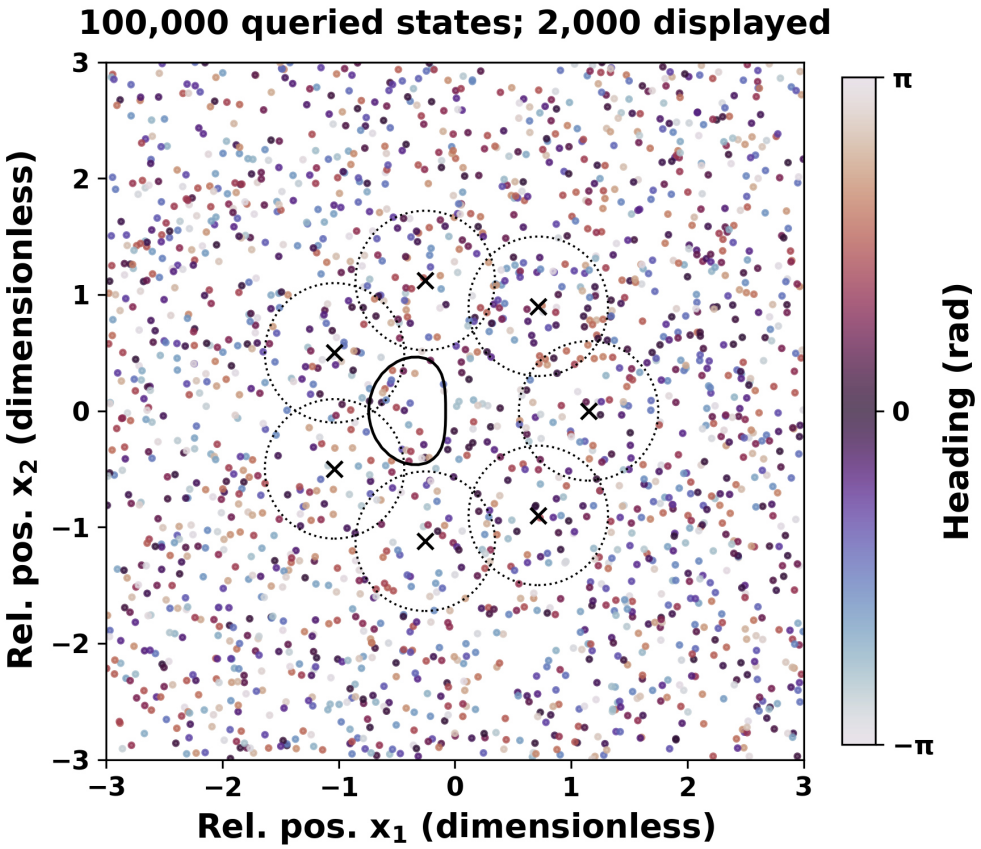


Figure 7: Phase-space view of queried bird states, colored by heading. Stars mark flock centers and dashed circles mark capture radii. The zero contour supplies geometric context for the displayed population; a value sign requires an error margin before it can be used as a safety certificate.

On the finest grid, the maximum full residual value errors are 1.07×10^{-8} in two dimensions and 3.00×10^{-8} in three dimensions. The independent finite-difference errors are 1.68×10^{-6} and 2.08×10^{-6}, respectively. These deterministic comparisons check the implemented identity and discretizations against a known solution; they do not establish uniform error bounds for the game Hamiltonians.

D. Limitations

The retained-source theory requires a positive invariant region and a norm that controls gradients. The contraction interval depends on source sensitivity, dimension, and viscosity. Long-horizon convergence requires renewed bounds on each interval. Signed-distance initial conditions and nonsmooth game Hamiltonians need regularity and positivity arguments beyond the bounded smooth diagnostic.

The Monte Carlo theorem controls a fixed finite collection of source-update values. It does not control gradients or reconstruction between queries. The full solve requires intermediate-time values and gradients at sampled states, so the size of an endpoint working array is not a bound for the retained-source method. Source quadrature, reconstruction, and accumulated update error must all be measured or bounded.

The deterministic implementation resolves periodic two- and three-dimensional fields with a fixed source-mode cutoff. Its grid storage grows with state dimension. The finite-difference

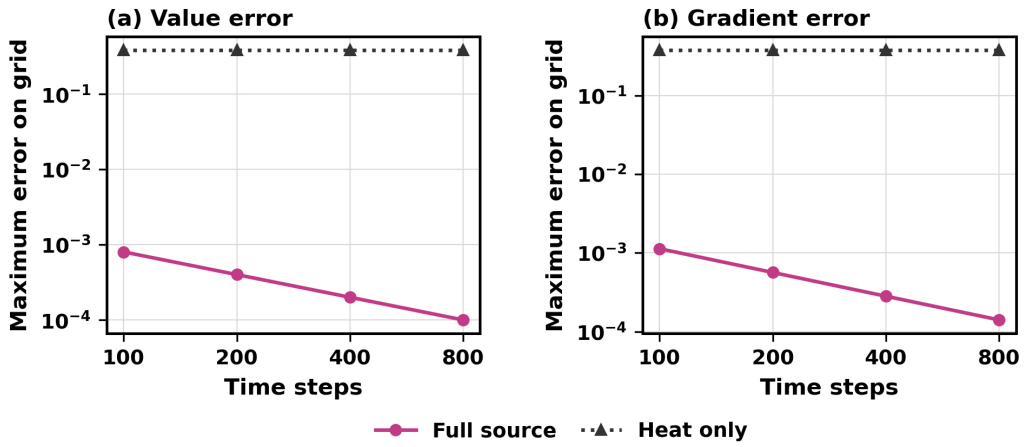


Figure 8: Temporal refinement of the full-source and homogeneous heat solutions. Maximum value error (left) and gradient error (right) at $t = 0.5$ on 128 periodic evaluation states. Circles denote the full source, and triangles denote homogeneous heat flow. Both axes are logarithmic; the state, value, and errors are dimensionless. The deterministic reference is the advected damped sine; no Monte Carlo averaging enters this comparison.

reference, sampled equation defects, and sensitivity experiments assess numerical consistency for the stated viscous problems. The box test combines domain, time-step, and coefficient changes. A uniform continuum defect bound remains a separate requirement. The supplied Gaussian source evaluator computes one prescribed-source update; closing the nonlinear stochastic source equation additionally requires intermediate-time reconstruction.

A positive approximate value is not a safety certificate without an error budget. Population-scale claims also require matching counts, solver configurations, and complete pipeline timing.

The Cauchy analysis does not solve reach–avoid obstacle complementarity, controlled stopping, or general multiplicative diffusion. A prescribed positive coefficient avoids dividing by a value gradient, but hard clipping can destroy the differentiability needed by the transformation. The source assumptions of global multilevel Picard complexity results must be verified separately [27].